Change-Point Detection in Tensor Graphical Models with Multiway Structure

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Abstract

Matrix- and tensor-variate data arise widely in modern applications, and detecting changes in their underlying dependence networks is important for understanding structural evolution. Existing methods often ignore the intrinsic multiway structure, resulting in severe statistical and computational challenges. To address these limitations, we propose a dynamic tensor Gaussian graphical model together with a separated change-point detection framework. The model allows different modes to remain stationary or change at coincident, distinct, or arbitrarily close locations, without imposing a common segmentation across modes. The proposed framework decomposes the joint estimation problem into mode-specific problems while leveraging information from the complementary modes. Such information aggregation increases the effective sample size and improves the effective signal-to-noise ratio for detecting structural changes in each mode. The separation also avoids a computationally intensive joint search over all mode-specific change points, permits concurrent implementation, and identifies the mode associated with each detected change. Under mild regularity conditions, we establish theoretical guarantees for mode-specific

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change-point localization, precision-matrix estimation, and graph recovery under a general class of folded-concave penalties, allowing the mode dimensions to grow with and even exceed the number of tensor observations. We further extend the framework to multiple change points through an efficient moving-window procedure. Numerical experiments demonstrate accurate recovery of mode-specific change points and evolving graphical structures with favorable computational performance.

Keywords: Tensor Gaussian graphical models, change-point detection, multiway structure, folded-concave penalties, graph recovery

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1 Introduction

High-dimensional data involving a large collection of variables are now routinely encountered in statistics and machine learning. In many applications, a central goal is to uncover the network of relationships among these variables rather than study their marginal behavior separately. Gaussian graphical models provide a principled framework for this purpose by encoding conditional dependence through the precision matrix, whose zero off-diagonal entries correspond to conditionally independent pairs of variables (Lauritzen 1996). A substantial literature has developed methods for high-dimensional Gaussian graphical model estimation and inference, including neighborhood-based, likelihood-based, multiple-testing, and constrained-estimation approaches (Meinshausen & Bühlmann 2006, Banerjee et al. 2008, Yuan & Lin 2007, Friedman et al. 2008, Xia et al. 2015, Liu 2013, Cai et al. 2011, 2016).

In many modern applications, however, each observation is naturally organized as a matrix or a higher-order tensor rather than an unstructured vector. For example, international trade observed over a given period can be represented by a country-by-commodity matrix, whereas a subject-level fMRI image forms a three-dimensional tensor indexed by spatial coordinates (Leng & Tang 2012, Lyu et al. 2020). Directly vectorizing such observations and applying a conventional Gaussian graphical model leads to a precision matrix whose dimension is the product of the mode dimensions. This results in a rapid growth in the number of parameters and hence a severe curse of dimensionality. More importantly, vectorization ignores the intrinsic multiway organization of the data and obscures the network structures associated with the individual modes. These considerations motivate graphical models that explicitly preserve the matrix or tensor structure and recover interpretable mode-specific dependence networks.

To accommodate the intrinsic multiway structure of such data, tensor graphical models have received increasing attention in recent years. A widely used formulation is based on the tensor normal distribution with a Kronecker-separable covariance structure. Specifically, let $\mathcal{X}_i \in \mathbb{R}^{p_1 \times \cdots \times p_M}$ be an order-M tensor observation satisfying $\mathcal{X}_i \sim \mathcal{TN}(\mathbf{0}; \mathbf{\Sigma}_1, \ldots, \mathbf{\Sigma}_M)$, or equivalently, $\text{vec}(\mathcal{X}_i) \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_M \otimes \cdots \otimes \mathbf{\Sigma}_1)$, where $\mathbf{\Sigma}_m \in \mathbb{R}^{p_m \times p_m}$ is the covariance factor associated with mode m. Let $\mathbf{\Omega}_m = \mathbf{\Sigma}_m^{-1}$ denote the corresponding mode-specific precision matrix. For any two distinct nodes $i_m, j_m \in [p_m]$, a zero off-diagonal entry $(\mathbf{\Omega}_m)_{i_m j_m} = 0$ indicates the absence of an edge between them in the mode-m conditional dependence graph. The precision matrix of $\text{vec}(\mathcal{X}_i)$ is given by $\mathbf{\Omega} = \mathbf{\Omega}_M \otimes \cdots \otimes \mathbf{\Omega}_1$, which combines the mode-specific networks to characterize the conditional independence structure of the entire tensor. In particular, for two tensor entries indexed by $\boldsymbol{i} = (i_1, \ldots, i_M)$ and $\boldsymbol{j} = (j_1, \ldots, j_M)$, under Gaussianity, these two tensor entries are conditionally independent given all the remaining entries if and only if $\prod_{m=1}^{M} (\mathbf{\Omega}_m)_{i_m j_m} = 0$, or equivalently, if $(\mathbf{\Omega}_m)_{i_m j_m} = 0$ for at least one mode m. Thus, each $\mathbf{\Omega}_m$ characterizes the network within a particular mode, while their Kronecker product determines the full conditional independence graph of the tensor.

A growing literature has developed statistical methods for estimating and analyzing these structured graphical models. For matrix-valued data, representative work includes penalized estimation of row and column precision matrices (Leng & Tang 2012, Zhou 2014), structural testing and differential network analysis (Xia & Li 2017, 2019, Chen & Liu 2019), and joint estimation of multiple matrix graphs (Huang & Chen 2015, Zhu & Li 2018, Liu et al. 2023). These ideas have further been extended to higher-order tensor graphical models by exploiting Kronecker-separable covariance or precision structures (Greenewald et al. 2019, Lyu et al. 2020, Min et al. 2022, Ren et al. 2024).

Most existing matrix and tensor graphical models assume that the mode-specific precision matrices remain invariant across observations. This assumption is reasonable for a homogeneous sample, but can be overly restrictive when tensor observations are collected over time or another ordered domain. For example, relationships among countries and commodities may be altered by trade-policy changes, geopolitical conflicts, or supply-chain disruptions; dependence among financial assets may shift after crises or market reorganizations; and environmental or biological networks may change following extreme events or across disease stages. Pooling observations from such heterogeneous regimes under a single static model may obscure newly formed or disappearing connections and yield a misleading representation of the underlying dependence structure. These considerations motivate a dynamic tensor graphical model in which each mode-specific precision matrix remains stable within regimes but changes abruptly at unknown locations.

Motivated by these applications, we consider a sequence of tensor observations $\{\mathcal{X}_i\}_{i=1}^n$, where $\mathcal{X}_i \in \mathbb{R}^{p_1 \times \cdots \times p_{\mathfrak{M}}}$ follows $\mathcal{X}_i \sim \mathcal{TN}\big(\mathbf{0}; \mathbf{\Sigma}_{1,i}^*, \ldots, \mathbf{\Sigma}_{\mathfrak{M},i}^*\big)$ or equivalently, $\mathrm{vec}(\mathcal{X}_i) \sim \mathcal{N}\big(\mathbf{0}, \mathbf{\Sigma}_{\mathfrak{M},i}^* \otimes \cdots \otimes \mathbf{\Sigma}_{1,i}^*\big)$. For each mode $m \in [\mathfrak{M}]$, let $\mathbf{\Omega}_{m,i}^* = (\mathbf{\Sigma}_{m,i}^*)^{-1}$ denote the mode-specific precision matrix at index i. We assume that $\mathbf{\Omega}_{m,i}^*$ is piecewise constant: there exist an unknown number K_m^* of change points $0 = \tau_{m,0}^* < \tau_{m,1}^* < \cdots < \tau_{m,K_m}^* < \tau_{m,K_m+1}^* = n$ such that

$$\mathbf{\Omega}_{m,i}^* = \mathbf{\Omega}_m^{(k)*}, \qquad \tau_{m,k-1}^* < i \leq \tau_{m,k}^*, \quad k = 1, \ldots, K_m^* + 1.$$

Thus, the conditional dependence graph within each mode remains stable between successive change points but may change abruptly across regimes. The model is flexible in allowing both the number and locations of change points to vary across modes. For example, when $M = 2$, modes 1 and 2 may each contain a single change point. These two change points may coincide, occur at distinct locations, or be different but arbitrarily close to one another. Alternatively, mode 1 may contain one change point while mode 2 contains two or more. Hence, the model imposes no common segmentation or cross-mode spacing requirement, and can accommodate simultaneous, asynchronous, and closely spaced mode-specific structural changes.

Change-point detection for graphical models has so far been developed mainly for vector-valued observations with only one mode. Fused-penalty methods estimate a sequence of piecewise-constant networks by encouraging sparsity in each graph and similarity between adjacent graphs (Kolar & Xing 2012, Gibberd & Nelson 2017, Gibberd & Roy 2021). Other work has considered computationally efficient estimation of a single change point (Bybee & Atchadé 2018), simultaneous change-point estimation and graph recovery through nodewise regressions (Liu et al. 2021), online monitoring of sparse precision matrices (Keshavarz et al. 2020), and graphical change-point detection under missing observations (Londschien et al. 2021). Broader high-dimensional segmentation frameworks also accommodate changes in precision matrices (Bai & Safikhani 2023, Li et al. 2023), while change-point estimation for general Markov random fields has been studied by Roy et al. (2017). These methods treat the precision matrix as a single high-dimensional structural object.

A natural strategy for detecting structural changes in tensor graphical models is to vectorize each tensor observation and then apply existing change-point methods for vector-valued Gaussian graphical models. This approach, however, faces several fundamental limitations. First, vectorization leads to a severe curse of dimensionality. Writing $P = \prod_{m=1}^{M} p_m$, an unrestricted precision matrix for the vectorized observation contains $P(P+1)/2$ free parameters. For example, when $M = 3$ and $p_1 = p_2 = p_3 = 10$, the vectorized observation has dimension $P = 1000$, so its precision matrix contains 500,500 distinct entries which is prohibitively large relative to the sample sizes typically available and makes reliable estimation extremely challenging. This dimensionality is particularly problematic in macroeconomic and other low-frequency applications, where structural changes are common but the number of observations is often inherently limited. Second, change-point detection further amplifies the computational burden because graphical-model estimation must be repeated over many candidate split points or candidate segments. In particular, local scanning typically requires separate graph estimates on the left, right. Repeatedly solving high-dimensional penalized precision-matrix problems can therefore be prohibitively expensive. Third, a change detected in the vectorized precision matrix reveals only that the overall dependence structure has changed, but does not identify which mode is responsible for the change. Finally, standard multiple-change-point methods usually impose a minimum spacing condition on successive changes. If two different modes change at distinct but nearby locations, vectorized procedures may merge them into a single change or fall outside the conditions required by existing algorithms and theories. These limitations call for a method that preserves the tensor structure and detects structural changes separately along each mode.

Motivated by these challenges, we develop a high-dimensional, mode-separable framework for detecting structural changes in tensor Gaussian graphical models, allowing the dimension p_m of an individual mode to exceed the number n of tensor observations. Our first contribution is a flexible dynamic tensor Gaussian graphical model in which different modes may undergo structural changes at different locations. We begin with the fundamental setting where each mode contains at most one change point, while allowing the mode-specific change points to be distinct, coincident, or arbitrarily close. The proposed formulation therefore identifies changes separately across modes without requiring a common segmentation or any cross-mode spacing condition.

Second, we develop a mode-separable conditional profile procedure that jointly estimates the change point and the pre- and post-change graphical structures for each mode. Although the mode-specific parameters are coupled through the Kronecker structure, we condition on provisional precision matrices from the complementary modes and reduce the original joint problem to a collection of one-dimensional profile searches. For each candidate location, the two regime-specific precision matrices are estimated using folded-concave penalization, and the change point is selected by the resulting profile loss. The mode-specific parameters are updated synchronously and can be computed in parallel. This procedure avoids joint optimization over all mode-specific change points and estimation of the full precision matrix of dimension $\prod_{m=1}^{M} p_m$, substantially reducing the computational burden while retaining a mode-specific interpretation of each detected change.

Third, we establish mode-wise high-dimensional theory for both change-point localization and graphical-structure estimation. The theoretical guarantees are derived separately for each mode and do not require the change-point locations, ordering, or spacing in the complementary modes to be specified. By borrowing information from the complementary modes, our formulation increases the effective sample size available for estimating mode m by a factor of p_{-m}. Compared with existing vector graphical-model change-point methods, this gain leads to a weaker signal-to-noise requirement and a faster localization rate, with the theoretical complexity depending only on the dimension and sparsity of the target mode rather than those of the full vectorized graph. We further derive estimation error bounds for the pre- and post-change precision matrices of each mode and establish consistent recovery of the corresponding sparse graphical structures. These estimators achieve the same convergence rates as under known segmentation, showing that estimating the change point does not reduce the accuracy of the precision-matrix estimators.

Lastly, we extend the proposed procedure to multiple change-point detection by combining it with a moving-window scheme. The resulting method allows the number and locations of change points to vary freely across modes, while retaining the computational efficiency of mode-wise estimation. Extensive simulations demonstrate that the method performs stably across a wide range of settings and can accurately localize multiple change points even when the number of tensor observations is very limited.

Notation. We collect here the conventions that are used throughout the paper; symbols with a more local scope are defined where they first appear. For a positive integer a, write $[a] = \{1, \ldots, a\}$. Throughout, $\mathcal{M}$ denotes the number of modes (or ways) of the data, a generic mode is indexed by $\mathfrak{m} \in [\mathfrak{M}]$, and $\mathfrak{j}$ is reserved for a mode other than $\mathfrak{m}$; the matrix-variate case corresponds to $\mathfrak{M} = 2$, in which case mode 1 and mode 2 are the row and the column mode, respectively. The mode-$\mathfrak{m}$ dimension is $p_{\mathfrak{m}}$, and we set $p = \prod_{\mathfrak{m}\in[\mathfrak{M}]} p_{\mathfrak{m}}$

and $p_{-\mathfrak{m}} = p/p_{\mathfrak{m}}$. For an order-$\mathfrak{M}$ tensor $\mathcal{X} \in \mathbb{R}^{p_1 \times \cdots \times p_{\mathfrak{M}}}$, $\lceil \mathcal{X} \rfloor_{(\mathfrak{m})} \in \mathbb{R}^{p_{\mathfrak{m}} \times p_{-\mathfrak{m}}}$ is its mode-$\mathfrak{m}$ unfolding, $\mathsf{Vec}(\mathcal{X}) \in \mathbb{R}^p$ is its vectorization, and $\mathbf{0}$ denotes a zero tensor, matrix or vector whose size is clear from the context. We use star-notation to denote true values, so that $\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}$ is the true mode-$\mathfrak{m}$ precision matrix of the ith observation. Finally, $\mathrm{Tr}(\cdot)$ is the trace, $|\cdot|$ the determinant, $\|\cdot\|_F$ the Frobenius norm, $\|\mathbf{A}\|_{1,\mathrm{off}} = \sum_{a \neq b} |[\mathbf{A}]_{a,b}|$ the off-diagonal ℓ_1 norm, $(x)_+ = \max(x, 0)$, and $\mathbb{I}(\cdot)$ the indicator function.

2 Dynamic Tensor Graphical Change-Point Model

This section introduces the proposed dynamic tensor graphical model. Section 2.1 reviews the tensor graphical structure and the associated identifiability convention. Section 2.2 formulates the mode-wise AMOC model, allowing different modes to change at distinct locations or remain stationary. Section 2.3 illustrates the matrix-variate case, where the two modes correspond to row and column networks have a change point.

2.1 Tensor Graphical Structure

Let $\mathcal{X} = ([\mathcal{X}]_{j_1,\ldots,j_{\mathfrak{M}}})_{j_{\mathfrak{m}} \in [p_{\mathfrak{m}}],\, \mathfrak{m} \in [\mathfrak{M}]} \in \mathbb{R}^{p_1 \times \cdots \times p_{\mathfrak{M}}}$ be an order-$\mathfrak{M}$ tensor following a zero-mean tensor normal distribution (Lyu et al. 2020, Min et al. 2022), denoted by $\mathcal{X} \sim \mathcal{TN}(\mathbf{0}; \boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_{\mathfrak{M}})$, where $\boldsymbol{\Sigma}_{\mathfrak{m}} \in \mathbb{R}^{p_{\mathfrak{m}} \times p_{\mathfrak{m}}}$ collects the dependence along mode $\mathfrak{m}$. Equivalently, $\mathcal{X}$ is tensor normal if and only if its vectorization is multivariate normal with a Kronecker-separable covariance structure,

$$\mathsf{Vec}(\mathcal{X}) \sim \mathcal{N}(\mathbf{0},\ \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma} = \bigotimes_{\mathfrak{m}=\mathfrak{M}}^{1} \boldsymbol{\Sigma}_{\mathfrak{m}} = \boldsymbol{\Sigma}_{\mathfrak{M}} \otimes \cdots \otimes \boldsymbol{\Sigma}_1, \tag{1}$$

and the density of $\mathcal{X}$ is $(2\pi)^{-p/2}|\boldsymbol{\Sigma}|^{-1/2}\mathrm{etr}\{-\frac{1}{2}\boldsymbol{\Sigma}^{-1}\,\mathsf{Vec}(\mathcal{X})\,\mathsf{Vec}(\mathcal{X})^{\top}\}$, where etr is the exponential of the trace operator. When $\mathfrak{M} = 2$ the tensor $\mathcal{X}$ is a matrix, which we write as $\mathbf{X} \in \mathbb{R}^{p_1 \times p_2}$, and (1) reduces to the matrix normal distribution $\mathbf{X} \sim \mathcal{MN}(\mathbf{0}; \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$

with density $(2\pi)^{-p/2}|\boldsymbol{\Sigma}_2 \otimes \boldsymbol{\Sigma}_1|^{-1/2}\text{etr}\{-\frac{1}{2}(\boldsymbol{\Sigma}_2 \otimes \boldsymbol{\Sigma}_1)^{-1} \mathsf{Vec}(\mathbf{X}) \mathsf{Vec}(\mathbf{X})^\top\}$. Conditional independence relationships are encoded, mode by mode, by the mode-$\mathfrak{m}$ precision matrix $\boldsymbol{\Omega}_\mathfrak{m} = \boldsymbol{\Sigma}_\mathfrak{m}^{-1}$. Specifically, a zero in the (a, b) entry of $\boldsymbol{\Omega}_\mathfrak{m}$ means that the ath and bth slices along mode $\mathfrak{m}$ are conditionally independent given the remaining ones. In the matrix case, $\boldsymbol{\Omega}_1$ and $\boldsymbol{\Omega}_2$ are the row and column precision matrices, and the associated graphs are the row and column networks.

Note that the decomposition in (1) is not unique where replacing $(\boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_\mathfrak{M})$ by $(c_1\boldsymbol{\Sigma}_1, \ldots, c_\mathfrak{M}\boldsymbol{\Sigma}_\mathfrak{M})$ leaves $\boldsymbol{\Sigma}$ unchanged whenever $\prod_\mathfrak{m} c_\mathfrak{m} = 1$. To resolve this scale indeterminacy we adopt the normalization $\|\boldsymbol{\Omega}_\mathfrak{m}\|_F = 1, \mathfrak{m} \in [\mathfrak{M}]$, with a single overall scale parameter separated from the normalized mode-specific covariance factors. Similar normalization conventions are commonly adopted in matrix and tensor graphical models (Lyu et al. 2020, Min et al. 2022). The above normalization does not alter the graph structure of any $\boldsymbol{\Omega}_\mathfrak{m}$, in the sense that an identical set of nonzero edges is shared before and after renormalization. We will impose this constraint in the estimation procedure of Section 3.

2.2 Mode-wise AMOC Model

Suppose we independently observe tensor samples $\mathcal{X}_1, \ldots, \mathcal{X}_n$ drawn from

$$\mathcal{X}_i \sim \mathcal{TN}\big(\mathbf{0}; \boldsymbol{\Sigma}^\star_{1,i}, \ldots, \boldsymbol{\Sigma}^\star_{\mathfrak{M},i}\big), \qquad i \in [n], \tag{2}$$

where for each mode $\mathfrak{m} \in [\mathfrak{M}]$ the mode-$\mathfrak{m}$ covariance matrix $\boldsymbol{\Sigma}^\star_{\mathfrak{m},i}$, with associated precision matrix $\boldsymbol{\Omega}^\star_{\mathfrak{m},i} = (\boldsymbol{\Sigma}^\star_{\mathfrak{m},i})^{-1}$, is piecewise stationary in $i \in [n]$. Specifically, each mode is allowed to carry its own change point $0 < \tau^\star_\mathfrak{m} < n$ with

$$\boldsymbol{\Omega}^\star_{\mathfrak{m},i} = \begin{cases} \boldsymbol{\Omega}^{(1)\star}_\mathfrak{m}, & i = 1, \ldots, \tau^\star_\mathfrak{m}, \\ \boldsymbol{\Omega}^{(2)\star}_\mathfrak{m}, & i = \tau^\star_\mathfrak{m} + 1, \ldots, n, \end{cases} \qquad \mathfrak{m} \in [\mathfrak{M}], \tag{3}$$

where the pre-change and post-change precision matrices $\mathbf{\Omega}_{\mathfrak{m}}^{(1)\star}, \mathbf{\Omega}_{\mathfrak{m}}^{(2)\star} \in \mathbb{R}^{p_{\mathfrak{m}} \times p_{\mathfrak{m}}}$ are symmetric and positive definite. The essential feature of (3) is that the change points $\tau_1^\star, \dots, \tau_{\mathfrak{M}}^\star$ need not coincide: structural breaks along different modes are triggered by the same index i, but may occur at different locations. We refer to (2)–(3) as the dynamic tensor Gaussian graphical model (DTGGM), and to its matrix-variate special case $\mathfrak{M} = 2$ as the dynamic matrix Gaussian graphical model (DMGGM). Since each mode carries at most one change point, we call this the mode-wise at-most-one-change (mode-wise AMOC) structure.

Substituting (3) into the Kronecker convention (**??**) exhibits the model as a regime-switching structure for the full precision matrix. Specifically, writing $\mathbf{k} = (k_1, \dots, k_{\mathfrak{M}}) \in \{1, 2\}^{\mathfrak{M}}$ for a vector of regime labels, then the Kronecker-structured precision matrix for time point i has the following form:

$$\mathbf{\Omega}_i^\star = \sum_{\mathbf{k} \in \{1,2\}^{\mathfrak{M}}} \left\{ \bigotimes_{\mathfrak{m}=\mathfrak{M}}^{1} \mathbf{\Omega}_{\mathfrak{m}}^{(k_{\mathfrak{m}})\star} \right\} \mathbb{I}_{i;\mathbf{k}}(\{\tau_{\mathfrak{m}}^\star\}), \qquad \mathbb{I}_{i;\mathbf{k}}(\{\tau_{\mathfrak{m}}^\star\}) = \prod_{\mathfrak{m}=1}^{\mathfrak{M}} \mathbb{I}\big(i \le \tau_{\mathfrak{m}}^\star\big)^{2-k_{\mathfrak{m}}} \mathbb{I}\big(i > \tau_{\mathfrak{m}}^\star\big)^{k_{\mathfrak{m}}-1}. \tag{4}$$

In the matrix case $\mathfrak{M} = 2$, expression (4) reads $\mathbf{\Omega}_i^\star = \{\mathbf{\Omega}_2^{(l)\star}\} \otimes \{\mathbf{\Omega}_1^{(k)\star}\} \mathbb{I}_{i;k,l}(\{\tau_{\mathfrak{m}}^\star\})$, with the four indicators

$$\begin{aligned} &\mathbb{I}_{i;1,1}(\{\tau_{\mathfrak{m}}^\star\}) = \mathbb{I}(i \le \tau_1^\star, i \le \tau_2^\star), \;\; \mathbb{I}_{i;2,1}(\{\tau_{\mathfrak{m}}^\star\}) = \mathbb{I}(i > \tau_1^\star, i \le \tau_2^\star), \\ &\mathbb{I}_{i;1,2}(\{\tau_{\mathfrak{m}}^\star\}) = \mathbb{I}(i \le \tau_1^\star, i > \tau_2^\star), \;\; \mathbb{I}_{i;2,2}(\{\tau_{\mathfrak{m}}^\star\}) = \mathbb{I}(i > \tau_1^\star, i > \tau_2^\star). \end{aligned} \tag{5}$$

Several implications of the representation in (4) are worth highlighting. **First**, although the summation in (4) runs over all $2^{\mathfrak{M}}$ possible regime-label vectors, at most $\mathfrak{M}+1$ distinct regimes can be realized along the ordered observation index. To see this, sort the mode-specific change points as $\tau_{(1)}^\star \le \cdots \le \tau_{(\mathfrak{M})}^\star$. The mode-wise AMOC formulation allows these change points to coincide and also permits some modes to remain stationary. If several modes change at the same location, their corresponding regime indicators switch simultaneously and generate only one common boundary in the full precision matrix. If mode $\mathfrak{m}$ is stationary, so that $\boldsymbol{\Omega}_{\mathfrak{m}}^{(1)\star} = \boldsymbol{\Omega}_{\mathfrak{m}}^{(2)\star}$, its nominal change point $\tau_{\mathfrak{m}}^{\star}$ does not induce any change in $\boldsymbol{\Omega}_i^{\star}$ and hence contributes no regime boundary. Consequently, the observation index $[n]$ is partitioned by the distinct change-point locations among the modes that actually undergo structural changes. Within each resulting interval, exactly one indicator $\mathbb{I}_{i;\mathbf{k}}(\{\tau_{\mathfrak{m}}^{\star}\})$ is active and $\boldsymbol{\Omega}_i^{\star}$ remains constant. The number of realized regimes is therefore one plus the number of such distinct change-point locations, and may be smaller than $\mathfrak{M}+1$.

For example, when $\mathfrak{M} = \mathfrak{2}$ and both modes change at distinct locations, at most three of the four regimes in (5) can occur. If $\tau_{\mathfrak{1}}^{\star} < \tau_{\mathfrak{2}}^{\star}$, then $\mathbb{I}_{i;2,1}(\{\tau_{\mathfrak{m}}^{\star}\})$ is active between the two change points, whereas $\mathbb{I}_{i;1,2}(\{\tau_{\mathfrak{m}}^{\star}\})$ is identically zero. Conversely, if $\tau_{\mathfrak{1}}^{\star} > \tau_{\mathfrak{2}}^{\star}$, then $\mathbb{I}_{i;1,2}(\{\tau_{\mathfrak{m}}^{\star}\})$ is active between the two change points, whereas $\mathbb{I}_{i;2,1}(\{\tau_{\mathfrak{m}}^{\star}\})$ is identically zero. When $\tau_{\mathfrak{1}}^{\star} = \tau_{\mathfrak{2}}^{\star}$, only the regimes represented by $\mathbb{I}_{i;1,1}(\{\tau_{\mathfrak{m}}^{\star}\})$ and $\mathbb{I}_{i;2,2}(\{\tau_{\mathfrak{m}}^{\star}\})$ are realized.

Second, viewing the sequence solely through the vectorized observations $\{\mathsf{Vec}(\mathcal{X}_i)\}_{i=1}^n$ yields a Gaussian graphical model with the full precision matrix $\boldsymbol{\Omega}_i^{\star} = \bigotimes_{\mathfrak{m}=\mathfrak{M}}^{\mathfrak{1}} \boldsymbol{\Omega}_{\mathfrak{m},i}^{\star}$. Although each mode contains at most one change point, the sequence $\{\boldsymbol{\Omega}_i^{\star}\}_{i=1}^n$ generally contains multiple change points whenever different modes undergo structural changes at distinct locations. Moreover, such change points may be arbitrarily close when they arise from different modes. A direct application of conventional multiple-change-point methods would therefore require the simultaneous localization of several potentially nearby changes in a high-dimensional full precision matrix. This task can be statistically challenging when the number of tensor observations is limited and may be incompatible with methods requiring sufficient separation between successive change points (Londschien et al. 2021, Liu et al. 2021).

More importantly, even if the change points of $\boldsymbol{\Omega}_i^{\star}$ could be detected accurately, an analysis based only on $\{\mathsf{Vec}(\mathcal{X}_i)\}_{i=1}^n$ would reveal changes in the overall conditional-dependence structure but would not directly identify which mode-specific precision matrix is responsible for each change. In contrast, the mode-wise formulation preserves the Kronecker structure and attributes each detected structural break to its corresponding mode.

2.3 An illustrative example

To make the mode-wise AMOC structure concrete, we display it in the matrix case, which is the largest $\mathfrak{M}$ for which the mode-wise precision matrices can be read directly off a heatmap; see Figure 1. Specifically, we generate the matrix-variate data $\{\mathbf{X}_i = (\boldsymbol{\Omega}^\star_{1,i})^{-1/2}\mathbf{Z}_i(\boldsymbol{\Omega}^\star_{2,i})^{-1/2}\}_{i=1}^n$ according to simulation setting (ii) in Section **??**, where $n = 100$, $(p_1, p_2) = (12, 10)$, and there is a single change point of the row precision matrix $\boldsymbol{\Omega}^\star_{1,i}$ at $\tau^\star_1 = 30$ and a single change point of the column precision matrix $\boldsymbol{\Omega}^\star_{2,i}$ at $\tau^\star_2 = 60$. Before the change, we set the row precision matrix and the column precision matrix to be the *hub* graph and the *band* graph, respectively. After the change, we randomly permute the last $\alpha = 4$ nodes in the row and column precision matrices, for nodes $\{1, \ldots, p_1\}$ and $\{1, \ldots, p_2\}$ respectively.

Figure 1 displays the row and column precision matrices and their corresponding Kronecker-structured full precision matrices over these three regimes. The example shows that, although each mode contains only one change point, their distinct locations induce multiple changes in the full precision matrix. Our objective is to recover the mode-specific change points $\tau^\star_1$ and $\tau^\star_2$, together with the corresponding pre- and post-change precision matrices.

3 Methods

This section develops the proposed estimation procedure. Section 3.1 first introduces an ideal joint profile formulation and explains the computational difficulty arising from the simultaneous search over all mode-specific change points. Section 3.2 then derives a mode-wise conditional profile estimator by treating the precision factors of the complementary modes as given. Finally, Section 3.3 combines these conditional updates into the complete change point detection procedure, in which all modes are updated iteratively using the

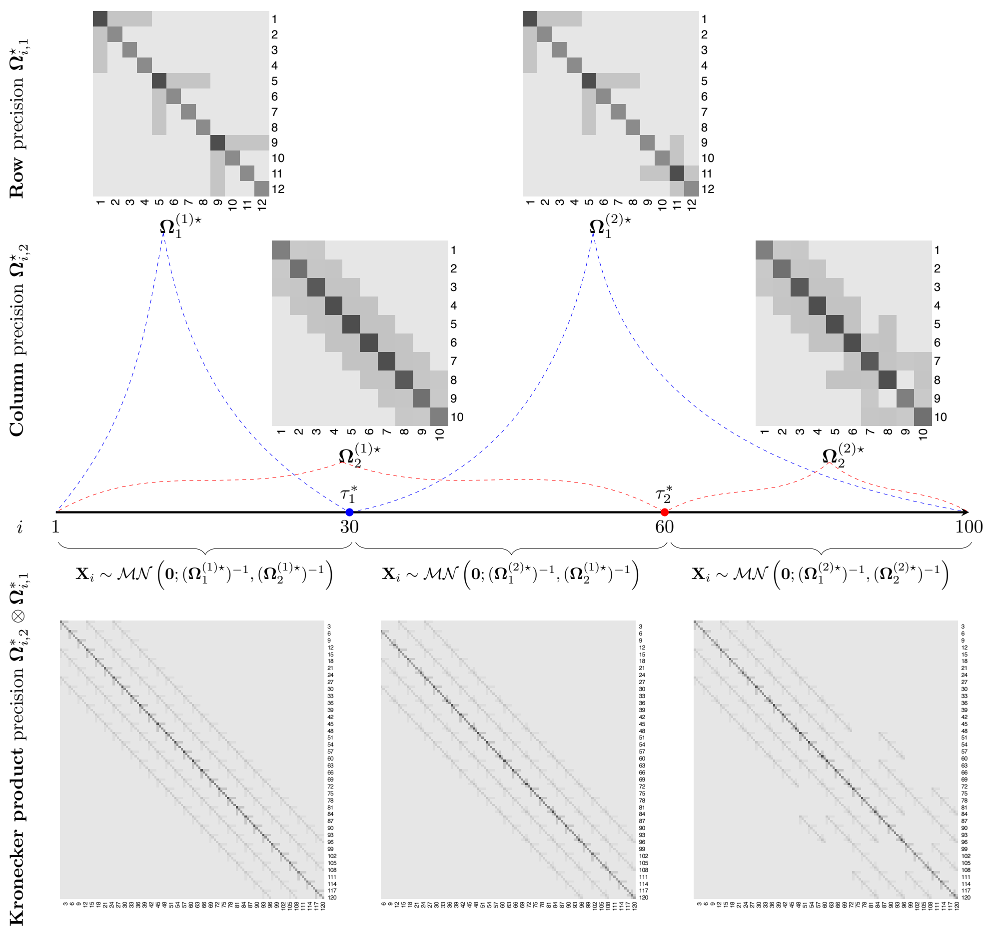

Figure 1: Heatmaps for networks of the mode-wise AMOC model with matrix-variate data.

complementary precision factors from the preceding iteration.

3.1 Joint Profile Formulation and Computational Challenge

Having specified the mode-wise AMOC model, we now turn to estimation. We first formulate an ideal joint estimator that simultaneously searches over all mode-specific change points and estimates the corresponding precision matrices. This formulation preserves the Kronecker structure of the full precision matrix, but, as shown below, requires optimization over an $\mathfrak{M}$-dimensional grid. Its main purpose is therefore to reveal the computational coupling across modes and motivate the separated estimation procedure developed in the next subsection.

For $k \in \{1, 2\}$, define $\boldsymbol{\theta}^{(k)} = \left(\boldsymbol{\Omega}_{\mathfrak{1}}^{(k)}, \ldots, \boldsymbol{\Omega}_{\mathfrak{M}}^{(k)}\right)$, where $\boldsymbol{\Omega}_{\mathfrak{m}}^{(1)}$ and $\boldsymbol{\Omega}_{\mathfrak{m}}^{(2)}$ denote the mode-$\mathfrak{m}$ precision matrices before and after its candidate change point, respectively. We collect all precision parameters as $\boldsymbol{\theta} = \left(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)}\right)$, and define the corresponding parameter space by

$$\boldsymbol{\Theta} = \left\{\boldsymbol{\theta} : \boldsymbol{\Omega}_{\mathfrak{m}}^{(k)} \in \mathbb{S}_{++}^{p_{\mathfrak{m}}}, \quad k \in \{1, 2\}, \quad \mathfrak{m} \in [\mathfrak{M}]\right\},$$

where $\mathbb{S}_{++}^{p_{\mathfrak{m}}} = \left\{\boldsymbol{A} \in \mathbb{R}^{p_{\mathfrak{m}} \times p_{\mathfrak{m}}} : \boldsymbol{A} = \boldsymbol{A}^\top \text{ and } \boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x} > 0 \text{ for all } \boldsymbol{x} \in \mathbb{R}^{p_{\mathfrak{m}}} \setminus \{\mathbf{0}\}\right\}$ denotes the set of symmetric positive-definite matrices of dimension $p_{\mathfrak{m}}$.

Let $\boldsymbol{\tau} = (\tau_{\mathfrak{1}}, \ldots, \tau_{\mathfrak{M}}) \in \mathcal{T} := \mathcal{T}_{\mathfrak{1}} \times \cdots \times \mathcal{T}_{\mathfrak{M}}$, where $\mathcal{T}_{\mathfrak{m}} \subset \{1, \ldots, n-1\}$ is the admissible candidate set for the change point of mode $\mathfrak{m}$. The components of $\boldsymbol{\tau}$ are attached to different modes rather than to their chronological order. Hence, they need not be ordered and may coincide across modes.

For a regime-label vector $\mathbf{k} = (k_{\mathfrak{1}}, \ldots, k_{\mathfrak{M}}) \in \{1, 2\}^{\mathfrak{M}}$, the mode-$\mathfrak{m}$ precision matrix is $\boldsymbol{\Omega}_{\mathfrak{m}}^{(k_{\mathfrak{m}})}$. Given $(\boldsymbol{\tau}, \boldsymbol{\theta}) \in \mathcal{T} \times \boldsymbol{\Theta}$, the normalized negative log-likelihood is defined as

$$\mathrm{L}_n(\boldsymbol{\tau}; \boldsymbol{\theta}) = \frac{1}{np} \sum_{i=1}^{n} \sum_{\mathbf{k} \in \{1,2\}^{\mathfrak{M}}} \left[\mathrm{Tr}\left\{\mathbf{S}_i \bigotimes_{\mathfrak{m}=\mathfrak{M}}^{\mathfrak{1}} \boldsymbol{\Omega}_{\mathfrak{m}}^{(k_{\mathfrak{m}})}\right\} - \log\left|\bigotimes_{\mathfrak{m}=\mathfrak{M}}^{\mathfrak{1}} \boldsymbol{\Omega}_{\mathfrak{m}}^{(k_{\mathfrak{m}})}\right|\right] \mathbb{I}_{i;\mathbf{k}}(\boldsymbol{\tau}), \tag{6}$$

where $\mathbf{S}_i = \mathsf{Vec}(\mathcal{X}_i)\,\mathsf{Vec}(\mathcal{X}_i)^\top$, and $\mathbb{I}_{i;\mathbf{k}}(\boldsymbol{\tau})$ is defined as in (4). The indicator selects the regime associated with observation i under the candidate vector $\boldsymbol{\tau}$. When $\mathfrak{M} = 2$, these indicators reduce to those in (5).

For illustration, consider the matrix-variate case $\mathfrak{M} = 2$. In this case, we have $\boldsymbol{\tau} = (\tau_1, \tau_2)$. Suppose now $\tau_1 < \tau_2$. Among the four possible regime labels in $\{1, 2\}^2$, only three can be realized along the ordered observation index. Specifically, we have

$$
\begin{aligned}
&\mathbb{I}_{i;(1,1)}(\boldsymbol{\tau}) = \mathbf{1}\{i \le \tau_1\}, \quad \mathbb{I}_{i;(2,1)}(\boldsymbol{\tau}) = \mathbf{1}\{\tau_1 < i \le \tau_2\}, \\
&\mathbb{I}_{i;(2,2)}(\boldsymbol{\tau}) = \mathbf{1}\{i > \tau_2\}, \quad \mathbb{I}_{i;(1,2)}(\boldsymbol{\tau}) = 0.
\end{aligned}
$$

Accordingly, the negative log-likelihood in (6) reduces to

$$
\begin{aligned}
\mathrm{L}_n(\boldsymbol{\tau};\boldsymbol{\theta}) = {} & \frac{1}{np}\sum_{i=1}^{\tau_1}\left[\mathrm{Tr}\left\{\mathbf{S}_i\left(\boldsymbol{\Omega}_2^{(1)} \otimes \boldsymbol{\Omega}_1^{(1)}\right)\right\} - \log\left|\boldsymbol{\Omega}_2^{(1)} \otimes \boldsymbol{\Omega}_1^{(1)}\right|\right] \\
& + \frac{1}{np}\sum_{i=\tau_1+1}^{\tau_2}\left[\mathrm{Tr}\left\{\mathbf{S}_i\left(\boldsymbol{\Omega}_2^{(1)} \otimes \boldsymbol{\Omega}_1^{(2)}\right)\right\} - \log\left|\boldsymbol{\Omega}_2^{(1)} \otimes \boldsymbol{\Omega}_1^{(2)}\right|\right] \\
& + \frac{1}{np}\sum_{i=\tau_2+1}^{n}\left[\mathrm{Tr}\left\{\mathbf{S}_i\left(\boldsymbol{\Omega}_2^{(2)} \otimes \boldsymbol{\Omega}_1^{(2)}\right)\right\} - \log\left|\boldsymbol{\Omega}_2^{(2)} \otimes \boldsymbol{\Omega}_1^{(2)}\right|\right].
\end{aligned}
$$

For any fixed $\boldsymbol{\tau} \in \mathcal{T}$, the precision parameters are estimated by

$$
\widehat{\boldsymbol{\theta}}(\boldsymbol{\tau}) = \left(\widehat{\boldsymbol{\theta}}^{(1)}(\boldsymbol{\tau}), \widehat{\boldsymbol{\theta}}^{(2)}(\boldsymbol{\tau})\right) \in \underset{\boldsymbol{\theta}\in\Theta}{\operatorname{argmin}}\left\{\mathrm{L}_n(\boldsymbol{\tau};\boldsymbol{\theta}) + \sum_{\mathfrak{m}=1}^{\mathfrak{M}}\left[\sqrt{\frac{\tau_{\mathfrak{m}}}{n}}\,\mathsf{P}_{\lambda_{\mathfrak{m}}}\left(\boldsymbol{\Omega}_{\mathfrak{m}}^{(1)}\right) + \sqrt{\frac{n-\tau_{\mathfrak{m}}}{n}}\,\mathsf{P}_{\lambda_{\mathfrak{m}}}\left(\boldsymbol{\Omega}_{\mathfrak{m}}^{(2)}\right)\right]\right\}. \tag{7}
$$

Here, $\mathsf{P}_{\lambda_{\mathfrak{m}}}(\cdot)$ is a sparsity-inducing penalty applied to the off-diagonal entries of a mode-$\mathfrak{m}$ precision matrix. The segment-dependent weights account for the different numbers of observations available before and after $\tau_{\mathfrak{m}}$.

The ideal joint profile estimator of the mode-specific change-point vector is then given by

$$
\widehat{\boldsymbol{\tau}} = (\widehat{\tau}_1, \ldots, \widehat{\tau}_{\mathfrak{M}}) \in \underset{\boldsymbol{\tau}\in\mathcal{T}}{\operatorname{argmin}}\,\mathrm{L}_n\big(\boldsymbol{\tau};\widehat{\boldsymbol{\theta}}(\boldsymbol{\tau})\big). \tag{8}
$$

For a fixed $\boldsymbol{\tau}$, problem (7) can be solved numerically by adapting the blockwise alternating algorithms developed for sparse matrix- and tensor-normal graphical models (Yin & Li 2012, Tsiligkaridis et al. 2013, Lyu et al. 2020).

The principal difficulty lies in the outer profile optimization in (8). Since $\boldsymbol{\tau}$ is an integer-valued vector, the profile loss $\mathrm{L}_n\big(\boldsymbol{\tau}; \widehat{\boldsymbol{\theta}}(\boldsymbol{\tau})\big)$ is discontinuous and piecewise constant over the Cartesian product $\mathcal{T} = \mathcal{T}_1 \times \cdots \times \mathcal{T}_{\mathfrak{M}}$. Consequently, gradient-based optimization is not applicable, and a direct implementation requires evaluating the profile criterion over all candidate vectors. When each mode has $O(n)$ admissible change-point locations, the number of candidate configurations is $|\mathcal{T}| = \prod_{\mathfrak{m}=1}^{\mathfrak{M}} |\mathcal{T}_{\mathfrak{m}}| = O(n^{\mathfrak{M}})$. More importantly, each evaluation of the profile loss $\mathrm{L}_n\big(\boldsymbol{\tau}; \widehat{\boldsymbol{\theta}}(\boldsymbol{\tau})\big)$ requires refitting all pre- and post-change mode-specific precision matrices in (7), **[which entails solving a nonconvex folded-concave penalized Kronecker graphical-model problem.]** If $\texttt{KronGGM}(n; p_1, \ldots, p_{\mathfrak{M}})$ denotes the computational cost of one such fit, exhaustive joint profiling has total complexity $O\left\{n^{\mathfrak{M}}\texttt{KronGGM}(n; p_1, \ldots, p_{\mathfrak{M}})\right\}$. Even in the matrix case $\mathfrak{M} = 2$, this requires an $O(n^2)$ outer search, with a complete graphical-model refit at every candidate pair. For $\mathfrak{M} \geq 3$, the number of candidate configurations grows polynomially in n with an exponent equal to the tensor order, making the joint search computationally prohibitive even for moderate sample sizes. Allowing multiple change points within each mode would further expand the search space and lead to an even more severe combinatorial burden.

The computational bottleneck therefore arises from jointly searching over the Cartesian product of all mode-specific candidate grids, with a complete refitting of the precision matrices required for each candidate vector. This observation motivates replacing the $\mathfrak{M}$-dimensional joint profile optimization by a sequence of mode-wise one-dimensional searches. We develop this separated estimation strategy in the next subsection.

3.2 Mode-wise Conditional Profile Estimation

The joint profile formulation in the preceding subsection reveals that the mode-specific change points and precision matrices are coupled through the Kronecker structure of the full precision matrix. At the same time, this structure suggests a natural conditional reduction. To see this, consider a fixed mode $\mathfrak{m} \in [\mathfrak{M}]$ and suppose, for the moment, that the precision matrices of all complementary modes are known at every observation index. That is, suppose that $\left\{\boldsymbol{\Omega}^{\star}_{\mathfrak{j},i} : \mathfrak{j} \in [\mathfrak{M}] \setminus \{\mathfrak{m}\},\ i \in [n]\right\}$ is available, and define $\boldsymbol{\Omega}^{\star}_{-\mathfrak{m},i} = \bigotimes_{\mathfrak{j}\neq\mathfrak{m}} \boldsymbol{\Omega}^{\star}_{\mathfrak{j},i}$ according to the Kronecker-product convention in (??).

Using these complementary precision matrices, each tensor observation can be whitened along all modes other than $\mathfrak{m}$. Specifically, define the corresponding whitened mode-$\mathfrak{m}$ matricization by

$$\mathbf{V}^{\star}_{\mathfrak{m},i} = \lceil \mathcal{X}_i \rfloor_{(\mathfrak{m})} \left(\boldsymbol{\Omega}^{\star}_{-\mathfrak{m},i}\right)^{1/2}.$$

Under the tensor-normal model, the $p_{-\mathfrak{m}}$ columns of $\mathbf{V}^{\star}_{\mathfrak{m},i}$ are independent Gaussian pseudo-observations with precision matrix $\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}$. Since $\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}$ follows the mode-wise AMOC structure in (3), these pseudo-observations satisfy a two-regime Gaussian graphical model with the single change point $\tau^{\star}_{\mathfrak{m}}$. Therefore, conditional on the precision matrices of the complementary modes, estimation along mode $\mathfrak{m}$ reduces to a standard one-dimensional single-change-point problem.

This oracle reduction provides the motivation for our **separated change-point detection** procedure, abbreviated as SCPD. In practice, the complementary precision matrices are unknown and we replace them by provisional estimates. These estimates are used to construct the corresponding whitened mode-$\mathfrak{m}$ matricizations and hence a mode-specific penalized profile criterion. Repeating this conditional construction across modes separates the original joint problem into a sequence of mode-wise change-point problems.

We now describe the conditional profile update for a fixed mode $\mathfrak{m} \in [\mathfrak{M}]$. Let

$$\widetilde{\mathbf{\Omega}}_{-\mathfrak{m}} := \left\{ \widetilde{\mathbf{\Omega}}_{\mathfrak{j},i} : \mathfrak{j} \in [\mathfrak{M}] \setminus \{\mathfrak{m}\},\ i \in [n] \right\}$$

denote a collection of provisional precision matrix for the complementary modes, where $\widetilde{\mathbf{\Omega}}_{\mathfrak{j},i} \in \mathbb{S}_{++}^{p_{\mathfrak{j}}}$. These matrices are allowed to vary with i because the complementary modes may have their own change points. Following the Kronecker-product convention, define

$$\widetilde{\mathbf{\Omega}}_{-\mathfrak{m},i} = \bigotimes_{\mathfrak{j} \neq \mathfrak{m}} \widetilde{\mathbf{\Omega}}_{\mathfrak{j},i}.$$

and define the corresponding whitened mode-$\mathfrak{m}$ matricization as $\widetilde{\mathbf{V}}_{\mathfrak{m},i} = \lceil \mathcal{X}_i \rfloor_{(\mathfrak{m})} \widetilde{\mathbf{\Omega}}_{-\mathfrak{m},i}^{1/2}$, and its transformed sample covariance matrix by $\widetilde{\mathbf{S}}_{\mathfrak{m},i} = \frac{1}{p_{-\mathfrak{m}}} \widetilde{\mathbf{V}}_{\mathfrak{m},i} \widetilde{\mathbf{V}}_{\mathfrak{m},i}^{\top}$. Under the true complementary precision factors, the $p_{-\mathfrak{m}}$ columns of $\widetilde{\mathbf{V}}_{\mathfrak{m},i}$ constitute the pseudo-observations associated with the ith tensor observation for mode $\mathfrak{m}$.

For a candidate change point $\tau \in \mathcal{T}_{\mathfrak{m}}$, let $\boldsymbol{\theta}_{\mathfrak{m}} = \left(\boldsymbol{\theta}_{\mathfrak{m}}^{(1)}, \boldsymbol{\theta}_{\mathfrak{m}}^{(2)} \right) \in \mathbf{\Theta}_{\mathfrak{m}}$ where $\mathbf{\Theta}_{\mathfrak{m}} = \left\{ \left(\boldsymbol{A}^{(1)}, \boldsymbol{A}^{(2)} \right) : \boldsymbol{A}^{(k)} \in \mathbb{S}_{++}^{p_{\mathfrak{m}}}, \quad k \in \{1,2\} \right\}$. Here, $\boldsymbol{\theta}_{\mathfrak{m}}^{(1)}$ and $\boldsymbol{\theta}_{\mathfrak{m}}^{(2)}$ represent the mode-$\mathfrak{m}$ precision matrices before and after the candidate change point, respectively. Define $\mathbb{I}_{i;1}(\tau) = \mathbb{I}(i \leq \tau)$ and $\mathbb{I}_{i;2}(\tau) = \mathbb{I}(i > \tau)$. The conditional mode-$\mathfrak{m}$ loss is then given by

$$\mathcal{L}_{n,\mathfrak{m}} \left(\tau, \boldsymbol{\theta}_{\mathfrak{m}}; \widetilde{\mathbf{\Omega}}_{-\mathfrak{m}} \right) = \frac{1}{n p_{\mathfrak{m}}} \sum_{i=1}^{n} \sum_{k=1}^{2} \mathbb{I}_{i;k}(\tau) \left[\mathrm{Tr} \left\{ \widetilde{\mathbf{S}}_{\mathfrak{m},i} \boldsymbol{\theta}_{\mathfrak{m}}^{(k)} \right\} - \log \left| \boldsymbol{\theta}_{\mathfrak{m}}^{(k)} \right| \right]. \tag{9}$$

For each fixed candidate $\tau \in \mathcal{T}_{\mathfrak{m}}$, the two mode-$\mathfrak{m}$ precision matrices are estimated by

$$\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}(\tau) = (\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(1)}(\tau), \widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(2)}(\tau)) \in \operatorname*{argmin}_{\boldsymbol{\theta}_{\mathfrak{m}} \in \mathbf{\Theta}_{\mathfrak{m}}} \left\{ \mathcal{L}_{n,\mathfrak{m}} \left(\tau, \boldsymbol{\theta}_{\mathfrak{m}}; \widetilde{\mathbf{\Omega}}_{-\mathfrak{m}} \right) + \sqrt{\frac{\tau}{n}} \mathsf{P}_{\lambda_{\mathfrak{m}}} \left(\boldsymbol{\theta}_{\mathfrak{m}}^{(1)} \right) + \sqrt{\frac{n-\tau}{n}} \mathsf{P}_{\lambda_{\mathfrak{m}}} \left(\boldsymbol{\theta}_{\mathfrak{m}}^{(2)} \right) \right\}. \tag{10}$$

Note that the two regime indicators have disjoint supports. Hence the optimization in (10) separates into two penalized graphical-model problems, one based on the observations before τ and the other based on those after τ.

After profiling out the precision parameters, based on $\{\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}(\tau)\}_{\tau \in \mathcal{T}_m}$, the mode-$\mathfrak{m}$ candidate change point is estimated by

$$\widehat{\tau}_{\mathfrak{m}} \in \operatorname*{argmin}_{\tau \in \mathcal{T}_{\mathfrak{m}}} \mathcal{L}_{n,\mathfrak{m}} \left(\tau, \widehat{\boldsymbol{\theta}}_{\mathfrak{m}}(\tau); \widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}} \right). \tag{11}$$

In this paper, we adopt a general class of folded-concave penalties for estimating the mode-specific precision matrices. For a symmetric matrix $\boldsymbol{A} = (a_{uv}) \in \mathbb{R}^{p_{\mathfrak{m}} \times p_{\mathfrak{m}}}$, we define

$$\mathsf{P}_{\lambda_{\mathfrak{m}}}(\boldsymbol{A}) = \sum_{u \neq v} \mathsf{p}_{\lambda_{\mathfrak{m}}}(|a_{uv}|),$$

where $\mathsf{p}_{\lambda_{\mathfrak{m}}}$ is applied to the off-diagonal entries. This class includes commonly used penalties such as SCAD (Fan & Li 2001) and MCP (Zhang 2010).

The use of folded-concave penalties is particularly motivated by the profile optimization over candidate change-point locations. When a candidate location τ differs from the true change point $\tau_{\mathfrak{m}}^{\star}$, one of the candidate segments generally contains observations from both regimes. A single precision matrix fitted to such mixed observations may represent a substantially denser intermediate structure than either of the two regime-specific precision matrices. Because the ℓ_1 penalty continues to shrink every nonzero entry, its accumulated shrinkage bias can be considerable in this setting, leading to inaccurate precision-matrix estimates and distorting the comparison of fitted losses across candidate locations. Folded-concave penalties mitigate this difficulty by promoting sparsity near the origin while imposing substantially less shrinkage on sufficiently large entries. Our theoretical analysis accommodates a general class of folded-concave penalties satisfying Assumptions 4 and 5. These conditions allow us to control the precision-matrix estimation error uniformly over candidate change-point locations and subsequently derive the localization rate of $\widehat{\tau}_{\mathfrak{m}}$.

3.3 Separated Change Point Detection

The complete SCPD procedure is summarized in Algorithm 1. Starting from an initial collection of mode- and index-specific precision factors, the algorithm updates each mode conditionally on the complementary precision factors obtained from the previous iteration. For mode $\mathfrak{m}$, these nuisance factors are used to construct the whitened mode-$\mathfrak{m}$ matricizations. At iteration t, for each candidate location τ, the two regime-specific precision matrices are first estimated by (10). The corresponding fitted losses are then compared through (11) to obtain $\tau_{\mathfrak{m}}^{(t)}$. The precision matrices associated with the selected location are retained as $\{(\boldsymbol{\Omega}_{\mathfrak{m}}^{(k)})^{(t)}\}_{k=1}^{2}$ and normalized according to the identifiability convention. They subsequently induce the updated sequence $\{(\boldsymbol{\Omega}_{\mathfrak{m},i})^{(t)}\}_{i=1}^{n}$.

Because every mode-specific update at iteration t depends only on quantities obtained at iteration $t-1$, the updates within the same iteration are independent of the updating order and can be implemented concurrently. Repeated iterations propagate the updated structural information across modes and may improve finite-sample performance. Although Algorithm 1 allows a user-specified number of iterations T, we show in Section **??** that the estimator obtained after the first complete iteration, $\tau_{\mathfrak{m}}^{(1)}$, already attains the stated localization rate. Subsequent iterations therefore serve as finite-sample refinements.

Remark 1 (Synchronized block updating)**.** *For each mode $\mathfrak{m}$, we treat $\left(\tau_{\mathfrak{m}}, \boldsymbol{\Omega}_{\mathfrak{m}}^{(1)}, \boldsymbol{\Omega}_{\mathfrak{m}}^{(2)}\right)$ as a single mode-specific parameter block. At iteration t, this block is updated conditionally on the parameter blocks of all complementary modes obtained at iteration $t-1$, through their implied index-specific precision factors. This scheme differs from the blockwise alternating updates commonly used in static matrix- and tensor-graphical estimation (Yin & Li 2012, Tsiligkaridis et al. 2013, Lyu et al. 2020). In those methods, the update of mode $\mathfrak{m}$ typically uses current-iteration estimates for the modes already visited and preceding-iteration estimates for the remaining modes. In contrast, SCPD updates every block $\left(\tau_{\mathfrak{m}}, \boldsymbol{\Omega}_{\mathfrak{m}}^{(1)}, \boldsymbol{\Omega}_{\mathfrak{m}}^{(2)}\right)$ using the same collection of preceding-iteration blocks. Thus, the mode-specific updates within an* *iteration do not depend on the updating order and can be performed concurrently.*

Algorithm 1 Iterative separated change-point detection across modes in tensor-variate data

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1: **Input:** tensor-variate samples $\mathcal{X}_1, \ldots, \mathcal{X}_n$, candidate sets $\{\mathcal{T}_{\mathfrak{m}}\}_{\mathfrak{m}=1}^{\mathfrak{M}}$, tuning parameters $\{\lambda_{\mathfrak{m}}\}_{\mathfrak{m}=1}^{\mathfrak{M}}$, and the maximum number of iterations T.

2: **Initialize** $\left\{(\boldsymbol{\Omega}_{\mathfrak{m},i})^{(0)}\right\}_{i=1}^{n}, \mathfrak{m} \in [\mathfrak{M}]$ as symmetric positive-definite matrices satisfying $\left\|(\boldsymbol{\Omega}_{\mathfrak{m},i})^{(0)}\right\|_F = 1, \mathfrak{m} \in [\mathfrak{M}], \quad i \in [n]$ and set $t = 0$.

3: **Repeat**:

4: $t = t + 1$.

5: **For** $\mathfrak{m} = \mathfrak{1}, \ldots, \mathfrak{M}$:

6: Let $\widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}}^{(t-1)} = \left\{\left\{(\boldsymbol{\Omega}_{\mathfrak{j},i})^{(t-1)}\right\}_{i=1}^{n} : \mathfrak{j} \in [\mathfrak{M}] \setminus \{\mathfrak{m}\}\right\}$.

7: For each $i \in [n]$, construct the complementary Kronecker precision matrix $\widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m},i}^{(t-1)} = \bigotimes_{\mathfrak{j}\neq\mathfrak{m}} (\boldsymbol{\Omega}_{\mathfrak{j},i})^{(t-1)}$ where the order of the Kronecker factors follows (**??**).

8: Construct the whitened mode-$\mathfrak{m}$ matricization $\widetilde{\mathbf{V}}_{\mathfrak{m},i}^{(t-1)} = \lceil \mathcal{X}_i \rfloor_{(\mathfrak{m})} \left(\widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m},i}^{(t-1)}\right)^{1/2}, i \in [n]$ and the corresponding transformed sample covariance matrices $\widetilde{\mathbf{S}}_{\mathfrak{m},i}^{(t-1)} = \frac{1}{p_{-\mathfrak{m}}} \widetilde{\mathbf{V}}_{\mathfrak{m},i}^{(t-1)} \left(\widetilde{\mathbf{V}}_{\mathfrak{m},i}^{(t-1)}\right)^{\top}$.

9: For each $\tau \in \mathcal{T}_{\mathfrak{m}}$, obtain $\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(t)}(\tau) = \left\{\left(\boldsymbol{\Omega}_{\mathfrak{m}}^{(k)}(\tau)\right)^{(t)}\right\}_{k=1}^{2}$ by solving (10) with $\widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}}$ replaced by $\widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}}^{(t-1)}$.

10: Obtain the mode-$\mathfrak{m}$ change-point estimate $\widehat{\tau}_{\mathfrak{m}}^{(t)} \in \operatorname{argmin}_{\tau\in\mathcal{T}_{\mathfrak{m}}} \mathcal{L}_{n,\mathfrak{m}}\left(\tau, \widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(t)}(\tau); \widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}}^{(t-1)}\right)$ according to (11), and set $\left\{\left(\boldsymbol{\Omega}_{\mathfrak{m}}^{(k)}\right)^{(t)}\right\}_{k=1}^{2} = \widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(t)}\left(\widehat{\tau}_{\mathfrak{m}}^{(t)}\right)$.

11: Define $(\boldsymbol{\Omega}_{\mathfrak{m},i})^{(t)} = \begin{cases} \left(\boldsymbol{\Omega}_{\mathfrak{m}}^{(1)}\right)^{(t)}, & i = 1, \ldots, \widehat{\tau}_{\mathfrak{m}}^{(t)}, \\ \left(\boldsymbol{\Omega}_{\mathfrak{m}}^{(2)}\right)^{(t)}, & i = \widehat{\tau}_{\mathfrak{m}}^{(t)} + 1, \ldots, n. \end{cases}$

12: Normalize the updated precision matrices according so that $\left\|(\boldsymbol{\Omega}_{\mathfrak{m},i})^{(t)}\right\|_F = 1, i \in [n]$.

13: **End for**

14: **Until** $t = T$.

15: **Output:** $\widehat{\boldsymbol{\tau}} = (\widehat{\tau}_{\mathfrak{1}}, \ldots, \widehat{\tau}_{\mathfrak{M}}) = \left(\widehat{\tau}_{\mathfrak{1}}^{(T)}, \ldots, \widehat{\tau}_{\mathfrak{M}}^{(T)}\right)$ and $\widehat{\boldsymbol{\theta}} = \left(\widehat{\boldsymbol{\theta}}^{(1)}, \widehat{\boldsymbol{\theta}}^{(2)}\right)$ where $\widehat{\boldsymbol{\theta}}^{(k)} = \left(\widehat{\boldsymbol{\Omega}}_{\mathfrak{1}}^{(k)}, \ldots, \widehat{\boldsymbol{\Omega}}_{\mathfrak{M}}^{(k)}\right) = \left(\left(\boldsymbol{\Omega}_{\mathfrak{1}}^{(k)}\right)^{(T)}, \ldots, \left(\boldsymbol{\Omega}_{\mathfrak{M}}^{(k)}\right)^{(T)}\right), \qquad k \in \{1, 2\}$.

---

4 Theory for Modes with a Structural Change

4.1 Assumptions

In this section, we establish the theoretical properties of the proposed SCPD procedure, including change-point localization, precision-matrix estimation, and graph recovery.

We first introduce the regularity conditions used in the subsequent analysis. Assumption 1 controls the mode-specific graph complexity, while Assumption 2 ensures that the covariance and precision matrices are well conditioned. Assumptions 3 regulate the growth of the model dimensions and the regularization level, respectively. Assumptions 4 and 5 characterize the local behavior of the folded-concave penalty and the separation of the true nonzero entries from zero. Finally, Assumption 6 requires the mode-specific structural change to dominate the effective high-dimensional noise level.

Assumption 1 (Sparsity). *For $i \in [n]$ and $\mathfrak{m} \in [\mathfrak{M}]$, define*

$$\mathbb{S}_{\mathfrak{m},i} = \left\{(a,b) : [\mathbf{\Omega}^{\star}_{\mathfrak{m},i}]_{a,b} \neq 0\right\},$$

and define the sparsity parameter $s_{\mathfrak{m},i} := |\mathbb{S}_{\mathfrak{m},i}| - p_{\mathfrak{m}}$, namely, the number of nonzero off-diagonal entries in $\mathbf{\Omega}^{\star}_{\mathfrak{m},i}$. Let $\mathbb{S}_{\mathfrak{m}} = \bigcup_{i=1}^{n} \mathbb{S}_{\mathfrak{m},i}$ and $s_{\mathfrak{m}} = |\mathbb{S}_{\mathfrak{m}}|$.

Assumption 2 (Bounded eigenvalues). *For any $i \in [n]$ and $\mathfrak{m} \in [\mathfrak{M}]$, there exists a constant $\rho > 0$ such that*

$$0 < \rho \leq \Lambda_{\min}(\mathbf{\Sigma}^{\star}_{\mathfrak{m},i}) \leq \Lambda_{\max}(\mathbf{\Sigma}^{\star}_{\mathfrak{m},i}) \leq 1/\rho < \infty,$$

where $\Lambda_{\min}(\cdot)$ and $\Lambda_{\max}(\cdot)$ denote the minimum and maximum eigenvalues, respectively.

Assumption 2 uniformly rules out degeneracy of the mode-specific covariance matrices. It guarantees stable inversion and implies the same uniform eigenvalue bounds for the corresponding precision matrices $\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}$. This condition also provides the local curvature required for controlling the precision-matrix estimation error.

Assumption 3 (Tuning). *For any* $\mathfrak{m} \in [\mathfrak{M}]$ *and some constant* $C_2 > 0$*, the tuning parameter* $\lambda_{\mathfrak{m}}$ *satisfies*

$$[\mathbf{C_2^{-1} p_{\mathfrak{m}}^{-1}} \sqrt{\frac{\log \mathbf{p_{\mathfrak{m}}}}{\mathbf{n p_{-\mathfrak{m}}}}} \le \lambda_{\mathfrak{m}} \le \mathbf{C_2 p_{\mathfrak{m}}^{-1}} \sqrt{\frac{\log \mathbf{p_{\mathfrak{m}}}}{\mathbf{n p_{-\mathfrak{m}}}}},]$$

and

$$p_{\mathfrak{m}}^{-2}(p_{\mathfrak{m}} + s_{\mathfrak{m}}) \log p_{\mathfrak{m}} / (n p_{-\mathfrak{m}}) = O(\lambda_{\mathfrak{m}}^2).$$

Assumption 4 (Folded-concave penalty). *The penalty function* $\mathsf{P}_\lambda(\cdot)$ *is nondecreasing and concave on* $[0, \infty)$ *with* $\mathsf{P}_\lambda(0) = 0$*, is twice differentiable on* $(0, \infty)$*, and is singular at the origin, with* $\lim_{x \downarrow 0} \mathsf{P}_\lambda(x)/(\lambda x) = \varpi > 0$*. Moreover, we require* $[\lim_{\mathbf{x} \to \infty} \mathsf{P}'_\lambda(\mathbf{x}) = \mathbf{0}.]$ *and there exist constants* C *such that, whenever* $x_1, x_2 > C\lambda$*,*

$$\left| \mathsf{P}''_\lambda(x_1) - \mathsf{P}''_\lambda(x_2) \right| \le D |x_1 - x_2|.$$

Assumption 5 (Signal separation). *Uniformly over* $i \in [n]$ *and* $\mathfrak{m} \in [\mathfrak{M}]$*, the nonzero off-diagonal entries of* $\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}$ *satisfy*

$$\max_{[\substack{(a,b) \in \mathbb{S}_{\mathfrak{m},i} \\ a \neq b}]} \mathsf{P}'_{\lambda_{\mathfrak{m}}} \left(\left| [\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}]_{a,b} \right| \right) = o \left[p_{\mathfrak{m}}^{-1} \left\{ 1 + \frac{p_{\mathfrak{m}}}{s_{\mathfrak{m}} + 1} \right\} \sqrt{\frac{\log p_{\mathfrak{m}}}{n p_{-\mathfrak{m}}}} \right],$$

$$\max_{\substack{(a,b) \in \mathbb{S}_{\mathfrak{m},i} \\ a \neq b}} \left| \mathsf{P}''_{\lambda_{\mathfrak{m}}} \left(\left| [\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}]_{a,b} \right| \right) \right| = o(p_{\mathfrak{m}}^{-1}),$$

and

$$\min_{\substack{(a,b) \in \mathbb{S}_{\mathfrak{m},i} \\ a \neq b}} \frac{\left| [\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}]_{a,b} \right|}{\lambda_{\mathfrak{m}}} \longrightarrow \infty, \qquad n \to \infty.$$

Assumptions 4 and 5 characterize the local behavior of the folded-concave penalty and the separation of the true nonzero entries from zero, respectively. The penalty class covers commonly used folded-concave penalties such as SCAD and MCP. These penalties behave similarly to the ℓ_1 penalty near the origin and hence retain its sparsity-inducing property, while imposing substantially less shrinkage on sufficiently large coefficients.

Conditions on the derivatives of a folded-concave penalty have been used in static graphical-model estimation for vector or matrix-valued graphical models (Lam & Fan (2009), Leng & Tang (2012)). The first derivative controls the shrinkage bias at the true nonzero entries, whereas the second derivative controls the local concavity of the penalized objective. The beta-min condition keeps the true nonzero entries sufficiently separated from the singular region near zero.

Remark 2. *In our dynamic multiway setting, these conditions additionally provide uniform control of the candidate-dependent population targets and their penalized estimators arising from the change-point scan. They therefore enter not only precision-matrix estimation and graph recovery, but also the comparison of candidate profile losses and the resulting change-point localization. Moreover, their stochastic scale is governed by the effective sample size* $np_{-\mathfrak{m}}$*, reflecting the information contributed by the complementary modes. This differs from existing static tensor graphical-model theory, which is primarily developed under* ℓ_1 *regularization (Lyu et al. 2020, Min et al. 2022).*

To ensure that the structural change in each mode can be distinguished from the high-dimensional estimation noise, we next impose a mode-specific signal-to-noise condition. For each $\mathfrak{m} \in [\mathfrak{M}]$, define the jump size by $\kappa_{\mathfrak{m}} := \left\| \boldsymbol{\Omega}_{\mathfrak{m}}^{(1)\star} - \boldsymbol{\Omega}_{\mathfrak{m}}^{(2)\star} \right\|_F$ and let $\delta_{\mathfrak{m}} := \min\{\tau_{\mathfrak{m}}^{\star}, n - \tau_{\mathfrak{m}}^{\star}\}$ denote the length of the shorter regime in mode $\mathfrak{m}$.

Assumption 6 (Signal strength)**.** *For each* $\mathfrak{m} \in [\mathfrak{M}]$, $\boldsymbol{\Omega}_{\mathfrak{m}}^{(1)\star} \neq \boldsymbol{\Omega}_{\mathfrak{m}}^{(2)\star}$*, so that* $\kappa_{\mathfrak{m}} > 0$*. There exists a sufficiently large constant* $C_{\mathrm{snr}} > 0$ *such that*

$$\delta_{\mathfrak{m}} \kappa_{\mathfrak{m}}^2 p_{-\mathfrak{m}} \geq C_{\mathrm{snr}} (s_{\mathfrak{m}} + p_{\mathfrak{m}}) \log p_{\mathfrak{m}}.$$

Assumption 6 requires the effective signal strength $\delta_{\mathfrak{m}} \kappa_{\mathfrak{m}}^2 p_{-\mathfrak{m}}$ to dominate the mode-specific estimation complexity $(s_{\mathfrak{m}} + p_{\mathfrak{m}}) \log p_{\mathfrak{m}}$. Compared with the signal-to-noise conditions in high-dimensional vector-valued graphical change-point models (Kaul et al. 2023). Assumption 6 has two distinctive features. First, the jump size and model complexity are defined at the mode level, involving $\kappa_{\mathfrak{m}}$, $p_{\mathfrak{m}}$, and $s_{\mathfrak{m}}$, rather than the dimension and sparsity of the full vectorized precision matrix. Second, the complementary modes enlarge the effective segment size from $\delta_{\mathfrak{m}}$ to $\delta_{\mathfrak{m}} p_{-\mathfrak{m}}$, thereby reducing the signal strength required for localization. Importantly, The complementary modes enter only through $p_{-\mathfrak{m}}$, regardless of whether they undergo structural changes. Consequently, the condition can accommodate $p_{\mathfrak{m}} > n$ when the information contributed by the complementary modes is sufficiently large.

4.2 Change-point Localization

Before presenting the main theoretical results, we introduce an additional notation to characterize the admissible provisional precision matrices used in the separated estimation procedure. For each $\mathfrak{m} \in [\mathfrak{M}]$ and $i \in [n]$, define

$$\mathbb{B}\big(\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}\big) = \big\{ \boldsymbol{\Omega}_i \in \mathbb{R}^{p_{\mathfrak{m}} \times p_{\mathfrak{m}}} : \boldsymbol{\Omega}_i = \boldsymbol{\Omega}_i^{\top},\ \boldsymbol{\Omega}_i \succ 0,\ \|\boldsymbol{\Omega}_i - \boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}\|_F \leq \varrho \big\}, \tag{12}$$

where $\varrho > 0$ is a fixed constant independent of the sample size and the mode dimensions.

The neighborhood in (12) is introduced to control the effect of replacing the unknown precision matrices of the complementary modes with provisional estimates. It prevents the corresponding transformation from excessively distorting the scale of the data and ensures that the transformed sample covariance matrices retain the concentration properties required for the subsequent analysis. These properties, in turn, keep the population objective well conditioned, provide stable local curvature, and preserve the separation of the profile criterion around the true change point.

Remark 3. *This condition is relatively mild. The radius ϱ is not required to shrink with the sample size and may be chosen as a large fixed constant. Hence, the provisional precision* *matrices need not be consistent estimators or converge to their population counterparts at any prescribed rate. This requirement is also compatible with common practice in matrix and tensor graphical modeling. Existing procedures typically initialize the mode-specific precision matrices using identity matrices (Leng & Tang 2012, Zhou 2014), randomly generated positive-definite matrices (Lyu et al. 2020), or data-driven covariance-based initializers Min et al. (2022). Although these initialization schemes do not by themselves guarantee membership in* (12)*, they suggest that accurate preliminary estimation is not generally required. Consistent with this observation, our numerical experiments show that the proposed SCPD procedure remains stable under relatively crude initializations.*

We are now ready to present the first main theoretical result, which characterizes the localization accuracy of the mode-specific change-point estimator from Algorithm 1.

Theorem 1. *For each $\mathfrak{m} \in [\mathfrak{M}]$, suppose there exists a constant $\varsigma > 1$ such that*

$$\frac{(p_{\mathfrak{m}} + s_{\mathfrak{m}})(\log p_{\mathfrak{m}})^{\varsigma}}{(\tau_{\mathfrak{m}}^{*} \wedge (n - \tau_{\mathfrak{m}}^{*}))p_{-\mathfrak{m}}} \to 0, \qquad n \to \infty. \tag{13}$$

Assume that Assumptions 1–6 hold. If the initialization satisfies $\widetilde{\mathbf{\Omega}}_{\mathfrak{j},i} \in \mathbb{B}(\mathbf{\Omega}^{\star}_{\mathfrak{j},i})$ for all $\mathfrak{j} \neq \mathfrak{m}$, then the mode-$\mathfrak{m}$ change-point estimator $\widehat{\tau}_{\mathfrak{m}}$ obtained from Algorithm 1 with $T = 1$ satisfies

$$\widehat{\tau}_{\mathfrak{m}} - \tau^{\star}_{\mathfrak{m}} = O_p\left(\frac{(s_{\mathfrak{m}} + p_{\mathfrak{m}})\log p_{\mathfrak{m}}}{p_{-\mathfrak{m}}\kappa_{\mathfrak{m}}^2}\right).$$

Theorem 1 reveals several features of the proposed estimator. First, its localization accuracy is determined by the mode-specific jump size $\kappa_{\mathfrak{m}}$, the graph complexity $(s_{\mathfrak{m}}+p_{\mathfrak{m}})\log p_{\mathfrak{m}}$, and the complementary-mode dimension $p_{-\mathfrak{m}}$. The rate shows that localization becomes more accurate when the mode-$\mathfrak{m}$ jump is stronger, the complementary modes provide more information, or the mode-$\mathfrak{m}$ graph is sparser. Second, the conventional localization rate for high-dimensional change-point models is typically governed by the graph complexity divided by the squared jump size (Lee et al. 2016, Kaul et al. 2023). Our rate contains the additional factor $p_{-\mathfrak{m}}$ in the denominator, showing that information from the complementary modes improves change-point localization. Third, the result is mode specific: provided that mode $\mathfrak{m}$ satisfies its own signal-to-noise condition, the displayed rate does not depend on the locations or signal strengths of change points in the other modes. Finally, although SCPD permits multiple iterative updates, the theorem shows that, conditional on the stated initialization requirement, a single mode-separation update is already sufficient to attain the localization rate.

Remark 4. *The condition (13) also allows the true change point to approach the data boundary. Unlike the usual requirement that the shorter segment contain at least an order of $s \log p$ observations, our condition is imposed on the effective segment size $\delta_{\mathfrak{m}} p_{-\mathfrak{m}}$. Hence, a sufficiently large $p_{-\mathfrak{m}}$ permits much shorter temporal segments and a search space extending close to the boundaries.*

4.3 Precision-matrix Estimation and Graph Recovery

We next study the estimation of the mode-specific precision matrices. We first establish an intermediate result for the pre-normalization estimators. To define their population targets, suppose that the true change point $\tau_{\mathfrak{m}}^{\star}$ is known and that the complementary-mode precision matrices $\widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}}$ are fixed. Define

$$\boldsymbol{\theta}_{\mathfrak{m}}^{\star} = \left\{ \boldsymbol{\theta}_{\mathfrak{m}}^{(1)\star}, \boldsymbol{\theta}_{\mathfrak{m}}^{(2)\star} \right\} \in \arg \min_{\boldsymbol{\theta}_{\mathfrak{m}} \in \Theta_{\mathfrak{m}}} \mathbb{E} \left[\mathcal{L}_{n,\mathfrak{m}} \left(\tau_{\mathfrak{m}}^{\star}, \boldsymbol{\theta}_{\mathfrak{m}}; \widetilde{\boldsymbol{\Omega}}_{-\mathfrak{m}} \right) \right].$$

where $\mathcal{L}_{n,\mathfrak{m}}$ is defined in (9). Note that the two segments induced by $\tau_{\mathfrak{m}}^{\star}$ are disjoint, $\boldsymbol{\theta}_{\mathfrak{m}}^{(1)\star}$ and $\boldsymbol{\theta}_{\mathfrak{m}}^{(2)\star}$ are the population minimizers associated with the pre-change and post-change regimes, respectively.

Theorem 2. *Under the conditions of Theorem 1, fix any $\mathfrak{m} \in [\mathfrak{M}]$. For each candidate change point $\tau \in \mathcal{T}_{\mathfrak{m}}$, let $\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}(\tau) = \left\{ \widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(1)}(\tau), \widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(2)}(\tau) \right\}$ denote the pre-normalization estimator* *obtained by solving* (10). *If* $\widetilde{\mathbf{\Omega}}_{\mathfrak{j},i} \in \mathbb{B}\left(\mathbf{\Omega}^{\star}_{\mathfrak{j},i}\right)$ *for all* $\mathfrak{j} \neq \mathfrak{m}$ *and* $i \in [n]$, *then, for each* $k \in \{1, 2\}$,

$$\left\|\widehat{\boldsymbol{\theta}}^{(k)}_{\mathfrak{m}}(\widehat{\tau}_{\mathfrak{m}}) - \boldsymbol{\theta}^{(k)\star}_{\mathfrak{m}}\right\|_F^2 = O_p\left\{\frac{(p_{\mathfrak{m}} + s_{\mathfrak{m}})\log p_{\mathfrak{m}}}{\delta_{\mathfrak{m}} p_{-\mathfrak{m}}}\right\}. \tag{14}$$

Theorem 2 provides an intermediate estimation result for the pre-normalization precision matrices. Although the estimators are evaluated at the estimated change point $\widehat{\tau}_{\mathfrak{m}}$, they attain the rate determined by the mode-specific complexity $(p_{\mathfrak{m}} + s_{\mathfrak{m}})\log p_{\mathfrak{m}}$ and the effective sample size $\delta_{\mathfrak{m}} p_{-\mathfrak{m}}$.

The preceding result concerns the pre-normalization population target, which may differ from the true precision matrix by a positive scaling factor. We next show that, after normalization, the resulting estimator converges directly to the identifiable mode-specific precision matrix at the same rate.

Theorem 3. *Under the conditions of Theorem 1, for any* $\mathfrak{m} \in [\mathfrak{M}]$ *and* $k \in \{1, 2\}$, *the normalized precision-matrix estimator produced by Algorithm 1 with* $T = 1$ *satisfies*

$$\left\|\widehat{\mathbf{\Omega}}^{(k)}_{\mathfrak{m}} - \mathbf{\Omega}^{(k)\star}_{\mathfrak{m}}\right\|_F^2 = O_p\left\{\frac{(p_{\mathfrak{m}} + s_{\mathfrak{m}})\log p_{\mathfrak{m}}}{\delta_{\mathfrak{m}} p_{-\mathfrak{m}}}\right\}. \tag{15}$$

Theorem 3 shows that normalization removes the scale ambiguity without affecting the convergence rate. Existing tensor graphical-model theory typically considers a static precision matrix and relies on ℓ_1-penalized estimation, yielding a squared Frobenius error of order $(p_{\mathfrak{m}} + s_{\mathfrak{m}})\log p_{\mathfrak{m}}/(np_{-\mathfrak{m}})$ (Lyu et al. 2020, Min et al. 2022). In contrast, our setting is dynamic and both the change point and the pre- and post-change precision matrices are unknown. Moreover, we employ a folded-concave penalty rather than an ℓ_1 penalty, thereby reducing the shrinkage bias for sufficiently large nonzero entries, particularly when candidate segments contain observations from different regimes.

Despite these additional difficulties, the normalized estimators evaluated at the change point $\widehat{\tau}_{\mathfrak{m}}$ achieve the corresponding static rate with n replaced by the shorter regime length

$\delta_{\mathfrak{m}}$. Thus, change-point uncertainty does not alter the first-order estimation rate, and both the pre- and post-change precision matrices can be estimated as accurately, in order, as if the true segmentation were known.

We next study the sparsity recovery of the estimated precision matrices. Since normalization preserves the zero pattern, it suffices to consider the pre-normalization estimators. The following theorem establishes their sparsistency.

Theorem 4 (Sparsistency). *Under the conditions of Theorem 1, fix any $\mathfrak{m} \in [\mathfrak{M}]$ and $k \in \{1,2\}$. Suppose that a local minimizer satisfies $\left\|\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(k)}(\widehat{\tau}_{\mathfrak{m}}) - \boldsymbol{\theta}_{\mathfrak{m}}^{(k)\star}\right\|_F^2 = O_p\left\{\frac{(p_{\mathfrak{m}}+s_{\mathfrak{m}})\log p_{\mathfrak{m}}}{\delta_{\mathfrak{m}} p_{-\mathfrak{m}}}\right\}$ and $\left\|\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(k)}(\widehat{\tau}_{\mathfrak{m}}) - \boldsymbol{\theta}_{\mathfrak{m}}^{(k)\star}\right\|_2^2 = O_p(\upsilon_{\mathfrak{m}})$ for some sequence $\upsilon_{\mathfrak{m}} \to 0$. If $[\frac{\log \mathbf{p}_{\mathfrak{m}}}{\delta_{\mathfrak{m}} \mathbf{p}_{-\mathfrak{m}}} + \upsilon_{\mathfrak{m}} = O\left(\frac{\mathbf{n}}{\delta_{\mathfrak{m}}}\mathbf{p}_{\mathfrak{m}}^{\mathbf{2}}\lambda_{\mathfrak{m}}^{\mathbf{2}}\right)]$ then, with probability tending to one, we have*

$$\left[\widehat{\boldsymbol{\theta}}_{\mathfrak{m}}^{(k)}(\widehat{\tau}_{\mathfrak{m}})\right]_{a,b} = 0 \qquad \textit{for all } (a,b) \in \mathbb{S}_{\mathfrak{m}}^c.$$

Theorem 4 shows that, despite the use of an estimated change point, the folded-concave estimator asymptotically removes all edges outside the true union support. Compared with existing static tensor graphical-model results based mainly on ℓ_1 penalties, our result applies to a dynamic setting with unknown pre- and post-change precision matrices.

5 Extension to Multiple Change-Point Detection

The preceding sections assume a mode-wise AMOC structure in (3), under which each mode contains at most one change point. In many applications, however, the graphical structure of a mode may change repeatedly. Let $\mathcal{T}_{\mathfrak{m}}^\star = \{\tau_{\mathfrak{m},1}^\star < \cdots < \tau_{\mathfrak{m},K_{\mathfrak{m}}}^\star\}$ denote the unknown change-point set of mode $\mathfrak{m}$, where $K_{\mathfrak{m}}$ is unknown and $K_{\mathfrak{m}} = 0$ corresponds to a stationary mode. Note that even if the $K_{\mathfrak{m}}$'s were known, a direct extension of the joint profile formulation in (8) would require searching over approximately $\prod_{\mathfrak{m}=1}^{\mathfrak{M}} \binom{n-1}{K_{\mathfrak{m}}}$ candidate segmentations, with a complete refitting of the Kronecker precision factors for each configuration. To overcome this difficulty, motivated by moving-window procedures for multiple change-point detection (Eichinger & Kirch (2018)), we keep the separation principle in Section 3 and replace the global search by an iterative mode-wise local scan.

Specifically, let G be a window width. To detect multiple structural changes in mode $\mathfrak{m}$, consider an admissible location $t_0 \in \{G, \ldots, n-G\}$. Define the adjacent windows $\mathcal{I}_{\mathrm{L}}(t_0) = \{t_0 - G + 1, \ldots, t_0\}$, $\mathcal{I}_{\mathrm{R}}(t_0) = \{t_0 + 1, \ldots, t_0 + G\}$, and let $\mathcal{I}(t_0) = \mathcal{I}_{\mathrm{L}}(t_0) \cup \mathcal{I}_{\mathrm{R}}(t_0)$. For any index set $\mathcal{I} \subseteq [n]$, define $\widehat{\mathbf{\Omega}}_{\mathfrak{m}}(\mathcal{I}) \in \operatorname{argmin}_{\mathbf{\Omega} \in \mathbb{S}_{++}^{p_{\mathfrak{m}}}} \mathcal{Q}_{\mathfrak{m}}(\mathcal{I}; \mathbf{\Omega})$, where $\mathcal{Q}_{\mathfrak{m}}(\mathcal{I}; \mathbf{\Omega})$ is the single-regime folded-concave penalized criterion restricted to the pseudo-samples $\{\widetilde{\mathbf{V}}_{\mathfrak{m},i}\}_{i \in \mathcal{I}}$. The corresponding minimized segment cost is $\ell_{\mathfrak{m}}(\mathcal{I}) = \mathcal{Q}_{\mathfrak{m}}\big(\mathcal{I}; \widehat{\mathbf{\Omega}}_{\mathfrak{m}}(\mathcal{I})\big)$.

We define the local gain statistic as

$$\mathcal{D}_{\mathfrak{m}}(t_0; G) = \ell_{\mathfrak{m}}\{\mathcal{I}(t_0)\} - \ell_{\mathfrak{m}}\{\mathcal{I}_{\mathrm{L}}(t_0)\} - \ell_{\mathfrak{m}}\{\mathcal{I}_{\mathrm{R}}(t_0)\}. \tag{16}$$

A large value of $\mathcal{D}_{\mathfrak{m}}(t_0; G)$ indicates that the two adjacent windows are better represented by different mode-$\mathfrak{m}$ precision matrices. We evaluate (16) over a grid $\mathcal{G}_G \subseteq \{G, \ldots, n-G\}$. Locations satisfying $\mathcal{D}_{\mathfrak{m}}(t_0; G) > \eta_{\mathfrak{m}}$ are first retained, after which only local maxima over neighborhoods of width proportional to G are kept. The surviving locations form the estimated change-point set

$$\widehat{\mathcal{T}}_{\mathfrak{m}} = \{\widehat{\tau}_{\mathfrak{m},1} < \cdots < \widehat{\tau}_{\mathfrak{m},\widehat{K}_{\mathfrak{m}}}\}.$$

If no location survives, mode $\mathfrak{m}$ is declared stationary. The estimated change points partition the observations into $\widehat{K}_{\mathfrak{m}} + 1$ segments. The resulting segment-wise estimates provide a piecewise stationary precision-matrix estimate for mode $\mathfrak{m}$.

In practice, the complementary precision factors required to construct the pseudo-samples are unknown. We therefore embed the above conditional procedure into the same synchronized iterative scheme used for the AMOC setting. At each iteration, every mode is updated using the complementary precision factors obtained from the preceding iteration.

The resulting change-point sets and segment-wise precision matrices are then propagated to the next iteration. The mode-specific updates within the same iteration can be performed concurrently. The complete implementation is summarized in Algorithm 2.

The window width G controls the trade-off between the stability of the local graphical-model fits and the ability to detect closely spaced changes. A distinctive feature of the present setting is that, when searching for change points in mode $\mathfrak{m}$, each observation index contributes $p_{-\mathfrak{m}}$ mode-$\mathfrak{m}$ pseudo-observations after the complementary modes are whitened. Consequently, a window containing G indices is effectively based on $Gp_{-\mathfrak{m}}$ pseudo-observations, rather than only G observations. When $p_{-\mathfrak{m}}$ is large, this additional information permits stable local precision-matrix estimation even with a relatively small window. Our numerical experiments confirm that modest values of G are generally sufficient to detect multiple change points accurately. This feature makes the moving-window construction particularly suitable for tensor-variate data, as it retains sensitivity to nearby changes without requiring large local windows.

Finally, we discuss the computational efficiency of the proposed procedure. In one iteration, the number of penalized graphical-model fits is $O\left\{\sum_{\mathfrak{m}=1}^{\mathfrak{M}}\big(|\mathcal{G}_G| + K_{\mathfrak{m}} + 1\big)\right\}$, where $|\mathcal{G}_G|$ corresponds to the moving-window scan and $K_{\mathfrak{m}} + 1$ to the segment-wise refitting. In addition, the mode-specific scans can be implemented concurrently. The procedure is therefore substantially faster than a joint search over all mode-specific segmentations. In our numerical experiments, a single iteration, corresponding to $T = 2$, already provides accurate change-point detection, while additional iterations are retained as optional finite-sample refinements.

Algorithm 2 Iterative SCPD for multiple change points in tensor-variate data

1: **Input:** tensor-variate observations $\mathcal{X}_1, \ldots, \mathcal{X}_n$, window width G, thresholds $\{\eta_{\mathfrak{m}}\}_{\mathfrak{m}=1}^{\mathfrak{M}}$, grid $\mathcal{G}_G$, and the maximum number of iterations T.

2: **Initialize** $\{(\mathbf{\Omega}_{\mathfrak{m},i})^{(0)}\}_{i=1}^{n}$, $\mathfrak{m} \in [\mathfrak{M}]$, as symmetric positive-definite matrices, and set $t = 0$.

3: **Repeat**:

4: $t = t + 1$.

5: **For** $\mathfrak{m} = \mathfrak{1}, \ldots, \mathfrak{M}$:

6: Construct the complementary precision factors $\widetilde{\mathbf{\Omega}}_{-\mathfrak{m}}^{(t-1)}$ from $\left\{(\mathbf{\Omega}_{\mathfrak{j},i})^{(t-1)} : \mathfrak{j} \in [\mathfrak{M}] \setminus \{\mathfrak{m}\},\ i \in [n]\right\}$.

7: Construct the mode-$\mathfrak{m}$ pseudo-samples $\{\widetilde{\mathbf{V}}_{\mathfrak{m},i}^{(t-1)}\}_{i=1}^{n}$ according to $\widetilde{\mathbf{V}}_{\mathfrak{m},i}^{(t-1)} = \lceil \mathcal{X}_i \rfloor_{(\mathfrak{m})} \left(\widetilde{\mathbf{\Omega}}_{-\mathfrak{m},i}^{(t-1)}\right)^{1/2}$, $i \in [n]$.

8: For each $t_0 \in \mathcal{G}_G$, fit the mode-$\mathfrak{m}$ precision matrices on $\mathcal{I}_{\mathrm{L}}(t_0)$, $\mathcal{I}_{\mathrm{R}}(t_0)$, and $\mathcal{I}(t_0)$, and compute the local gain statistic $\mathcal{D}_{\mathfrak{m}}^{(t)}(t_0; G)$ according to (16).

9: Retain the locations satisfying $\mathcal{D}_{\mathfrak{m}}^{(t)}(s; G) > \eta_{\mathfrak{m}}$ and select their local maxima to obtain $\mathcal{T}_{\mathfrak{m}}^{(t)} = \left\{\tau_{\mathfrak{m},1}^{(t)} < \cdots < \tau_{\mathfrak{m},K_{\mathfrak{m}}^{(t)}}^{(t)}\right\}$.

10: Refit one mode-$\mathfrak{m}$ precision matrix on each segment induced by $\mathcal{T}_{\mathfrak{m}}^{(t)}$, and use the resulting segment-wise estimates to construct $\{(\mathbf{\Omega}_{\mathfrak{m},i})^{(t)}\}_{i=1}^{n}$.

11: **End for**

12: **Until** $t = T$.

13: **Output:** $\widehat{\mathcal{T}}_{\mathfrak{m}} = \mathcal{T}_{\mathfrak{m}}^{(T)}$, $\mathfrak{m} \in [\mathfrak{M}]$ together with the corresponding segment-wise precision-matrix estimators.

6 [Numerical experiments]

In this section, we investigate the finite-sample performance of the proposed SCPD procedure. We first describe the data-generating mechanism, graph structures, and evaluation criteria, and then report simulation results under several representative scenarios. Throughout, we compare SCPD with two representative high-dimensional vector Gaussian graphical-model change-point methods: the approximate MM algorithm of Bybee & Atchadé (2018) and the binary-segmentation-based method of Londschien et al. (2021). Since these competing methods are designed for vector-valued observations, matrix data are vectorized before being passed to the competitors.

6.1 Simulation setup and evaluation metrics

Data generation. We consider a change-point model for the mode-specific precision matrix. Let the pre-change and post-change precision matrices be denoted by $\mathbf{\Omega}$ and $\mathbf{\Omega}(\alpha)$, respectively, both in $\mathbb{R}^{p\times p}$. For regime index $k \in \{1, 2\}$, write $\mathbf{\Omega}^{(1)\star} = \mathbf{\Omega}$ and $\mathbf{\Omega}^{(2)\star} = \mathbf{\Omega}(\alpha)$. To create structural changes, we first define the ordered index set $\boldsymbol{\pi}_1 = \{1, \ldots, p\}$ and then construct a partial permutation $\boldsymbol{\pi}_2 = \{\pi_1, \ldots, \pi_p\}$, where the first $p - \alpha$ indices remain unchanged while the last α indices are randomly permuted. The post-change precision matrix is then given by $\mathbf{\Omega}(\alpha) = \mathbf{P}\mathbf{\Omega}\mathbf{P}^\top$, where $\mathbf{P}$ is the permutation matrix induced by $\boldsymbol{\pi}_2$. This design changes the connectivity pattern of only the last α nodes and therefore controls the structural discrepancy between $\mathbf{\Omega}$ and $\mathbf{\Omega}(\alpha)$.

We consider three baseline graph structures for $\mathbf{\Omega}$, namely Band, Hub, and Cluster, and generate $\mathbf{\Omega}(\alpha)$ through the permutation mechanism above. After enforcing positive definiteness, all precision matrices are normalized by their Frobenius norm.

- **Band graph.** Construct $\mathbf{\Omega} = (\omega_{ij}) \in \mathbb{R}^{p\times p}$ with $\omega_{i,i+1} = \omega_{i+1,i} = 0.3$ and all other off-diagonal entries equal to 0. To guarantee positive definiteness, define

$$\mathbf{\Omega}^{\mathsf{band}} = \mathbf{\Omega} + \big(|\Lambda_{\min}(\mathbf{\Omega})| + 0.05\big)\mathbf{I}_p,$$

  and then standardize it to have unit diagonal.

- **Hub graph.** Partition the indices into five disjoint groups.[1] Let the first index in

[1] If p is not divisible by 5, we partition the first $5\lfloor p/5\rfloor$ indices evenly into five groups and assign the remaining indices sequentially to the last group. The same convention is used for the Cluster structure.

each group serve as a hub node and connect it to all remaining nodes in that group with edge weight 0.3. Denote the resulting matrix by $\mathbf{\Omega}$, then define

$$\mathbf{\Omega}^{\mathsf{hub}} = \mathbf{\Omega} + \big(|\Lambda_{\min}(\mathbf{\Omega})| + 0.05\big)\mathbf{I}_p,$$

followed by unit-diagonal standardization.

- [Cluster graph.] Again partition the indices into five disjoint groups. Within each group, set every off-diagonal entry equal to 0.2 and set all between-group off-diagonal entries to 0. Denote the resulting matrix by $\mathbf{\Omega}$, then define

$$\mathbf{\Omega}^{\mathsf{cluster}} = \mathbf{\Omega} + \big(|\Lambda_{\min}(\mathbf{\Omega})| + 0.05\big)\mathbf{I}_p,$$

and standardize it to have unit diagonal.

6.2 Single change point with $\mathfrak{M} = 2$

6.3 Single change point with $\mathfrak{M} = 3$

All data are generated under the two-way dynamic matrix graphical-model formulation. Once (n, p_1, p_2), the change-point locations $(\tau_1^\star, \tau_2^\star)$, and the row/column precision matrices $\{\mathbf{\Omega}_1^{(k)\star}\}_{k=1}^2$ and $\{\mathbf{\Omega}_2^{(k)\star}\}_{k=1}^2$ are specified, the data-generating mechanism is fully determined. We consider the following three settings.

(D1) Fix $n = 100$ and $(\tau_1^\star, \tau_2^\star) = (50, 50)$. The row and column precision matrices are

$$\mathbf{\Omega}_{1,i}^\star = \begin{cases} \mathbf{\Omega}^{\mathsf{hub}}, & i = 1, \dots, 50, \\ \mathbf{\Omega}^{\mathsf{hub}}(\alpha), & i = 51, \dots, 100, \end{cases} \qquad \mathbf{\Omega}_{2,i}^\star = \begin{cases} \mathbf{\Omega}^{\mathsf{band}}, & i = 1, \dots, 50, \\ \mathbf{\Omega}^{\mathsf{band}}(\alpha), & i = 51, \dots, 100. \end{cases}$$

(D2) **[Fix $n = 100$ and $(\tau_1^\star, \tau_2^\star) = (30, 60)$. The row and column precision matrices are]**

$$\boldsymbol{\Omega}_{1,i}^\star = \begin{cases} \boldsymbol{\Omega}^{\mathsf{cluster}}, & i = 1, \ldots, 30, \\ \boldsymbol{\Omega}^{\mathsf{cluster}}(\alpha), & i = 31, \ldots, 100, \end{cases} \qquad \boldsymbol{\Omega}_{2,i}^\star = \begin{cases} \boldsymbol{\Omega}^{\mathsf{band}}, & i = 1, \ldots, 60, \\ \boldsymbol{\Omega}^{\mathsf{band}}(\alpha), & i = 61, \ldots, 100. \end{cases}$$

(D3) Fix $n = 50$ and $(\tau_1^\star, \tau_2^\star) = (20, 25)$. The row and column precision matrices are

$$\boldsymbol{\Omega}_{1,i}^\star = \begin{cases} \boldsymbol{\Omega}^{\mathsf{hub}}, & i = 1, \ldots, 20, \\ \boldsymbol{\Omega}^{\mathsf{hub}}(\alpha), & i = 21, \ldots, 50, \end{cases} \qquad \boldsymbol{\Omega}_{2,i}^\star = \begin{cases} \boldsymbol{\Omega}^{\mathsf{band}}, & i = 1, \ldots, 25, \\ \boldsymbol{\Omega}^{\mathsf{band}}(\alpha), & i = 26, \ldots, 50. \end{cases}$$

Evaluation metrics. Because SCPD estimates the change-points in the two modes separately, we evaluate performance from both mode-wise and joint perspectives. For each mode $\mathfrak{m} \in \{1, 2\}$, we report the empirical mean and standard deviation of the estimated change-point $\widehat{\tau}_{\mathfrak{m}}$ over 100 Monte Carlo replications. To assess the joint accuracy of $(\widehat{\tau}_1, \widehat{\tau}_2)$, we use the (unnormalized) Hausdorff distance between the estimated and true change-point sets. In addition, we report the average estimated number of change-points. For SCPD, if $\widehat{\tau}_1 = \widehat{\tau}_2$, then the two estimates are treated as a single change-point when counting the number of detected changes.

6.4 Simulation results

Tables 1–3 summarize the simulation results under mechanisms **(D1)**–**(D3)**. We first examine the mode-wise localization accuracy. Across all three settings, the estimated change-points produced by SCPD are tightly concentrated around the true locations, with negligible bias and generally small standard deviations. This pattern is visible not only in the symmetric-change scenario **(D1)**, but also in the asynchronous settings **(D2)** and **(D3)**. Moreover, as the dimensions increase, the variability of the SCPD estimators tends

Table 1: Comparison of change-point detection performance under data-generating mechanism **(D1)**.

Method	$\widetilde{\mathbf{V}}_{\mathfrak{m},i}$ [a]	$p_1 = 15(6)$ [b] $p_2 = 15(6)$	$p_1 = 30(8)$ $p_2 = 15(6)$	$p_1 = 30(8)$ $p_2 = 30(8)$	$p_1 = 60(10)$ $p_2 = 60(10)$
SCPD (mode-1)	$\widetilde{\mathbf{\Omega}}_{2,i}$	49.98 (0.32) [c]	50.04 (0.42)	49.99 (0.26)	50.00 (0.00)
	$\mathbf{\Omega}^{\star}_{2,i}$	50.01 (0.33)	50.02 (0.20)	50.02 (0.20)	50.00 (0.00)
SCPD (mode-2)	$\widetilde{\mathbf{\Omega}}_{1,i}$	50.02 (0.72)	49.99 (0.10)	50.01 (0.10)	50.00 (0.00)
	$\mathbf{\Omega}^{\star}_{1,i}$	49.99 (0.69)	49.99 (0.10)	50.00 (0.00)	50.01 (0.10)
SCPD (two mode)	$\{\widetilde{\mathbf{\Omega}}_{\mathfrak{m},i}\}_{\mathfrak{m}=1}^{2}$	0.32|1.00 [d]	0.15|1.00	0.06|1.00	0.02|1.00
	$\{\mathbf{\Omega}^{\star}_{\mathfrak{m},i}\}_{\mathfrak{m}=1}^{2}$	0.26|1.00	0.05|1.00	0.04|1.00	0.01|1.00
hdcd	–	44.21 | 10.21	49.27|12.37	– [e]	–
BA	–	20.38|5.24	29.30|6.73	–	–

Notes: a. The second column specifies whether the auxiliary matrix in $\widetilde{\mathbf{V}}_{\mathfrak{m},i} = \lfloor \mathbf{X}_i \rceil_{(\mathfrak{m})} (\widetilde{\mathbf{\Omega}}_{-\mathfrak{m},i})^{1/2}$ is randomly initialized or set to the true value; b. The number in parentheses after p_m is the value of α, namely, the number of permuted nodes; c. For the mode-wise SCPD results, each entry reports the sample mean of the estimated change-point, with its standard deviation in parentheses; d. For the two-mode results, the quantity to the left of "—" is the unnormalized Hausdorff distance, while the quantity to the right is the average estimated number of change-points; e. When $p_1 \times p_2 = 30 \times 30$ or 60×60, the competing vectorized methods are numerically unstable due to the high dimensionality and limited sample size.

Table 2: Comparison of change-point detection performance under data-generating mechanism **(D2)**.

Method	$\widetilde{\mathbf{V}}_{\mathfrak{m},i}$	$p_1 = 12(4)$ $p_2 = 10(4)$	$p_1 = 40(6)$ $p_2 = 10(4)$	$p_1 = 40(6)$ $p_2 = 40(6)$	$p_1 = 40(6)$ $p_2 = 80(8)$
SCPD (mode-1)	$\widetilde{\mathbf{\Omega}}_{2,i}$	30.10 (0.58)	29.88 (1.07)	30.03 (0.22)	30.00 (0.00)
	$\mathbf{\Omega}^{\star}_{2,i}$	30.00 (0.47)	29.88 (0.86)	29.99 (0.10)	30.00 (0.00)
SCPD (mode-2)	$\widetilde{\mathbf{\Omega}}_{1,i}$	60.22 (1.28)	59.97 (0.30)	59.95 (0.39)	60.01 (0.22)
	$\mathbf{\Omega}^{\star}_{1,i}$	60.07 (0.90)	59.99 (0.17)	60.00 (0.28)	59.98 (0.24)
SCPD (two mode)	$\{\widetilde{\mathbf{\Omega}}_{\mathfrak{m},i}\}_{\mathfrak{m}=1}^{2}$	0.73|2.00	0.55|2.00	0.12|2.00	0.05|2.00
	$\{\mathbf{\Omega}^{\star}_{\mathfrak{m},i}\}_{\mathfrak{m}=1}^{2}$	0.48|2.00	0.39|2.00	0.07|2.00	0.04|2.00

Table 3: Comparison of change-point detection performance under data-generating mechanism **(D3)**.

Method	$\widetilde{\mathbf{V}}_{\mathfrak{m},i}$	$p_1 = 20(8)$ $p_2 = 20(8)$	$p_1 = 40(8)$ $p_2 = 20(8)$	$p_1 = 40(8)$ $p_2 = 40(8)$	$p_1 = 80(8)$ $p_2 = 40(8)$
SCPD (mode-1)	$\widetilde{\mathbf{\Omega}}_{2,i}$	20.00 (0.29)	19.84 (2.10)	20.04 (0.43)	20.07 (0.64)
	$\mathbf{\Omega}^{\star}_{2,i}$	20.00 (0.20)	20.05 (0.33)	20.01 (0.14)	20.05 (0.44)
SCPD (mode-2)	$\widetilde{\mathbf{\Omega}}_{1,i}$	25.04 (0.43)	24.99 (0.10)	24.97 (0.18)	24.74 (2.50)
	$\mathbf{\Omega}^{\star}_{1,i}$	25.01 (0.10)	24.99 (0.10)	25.00 (0.00)	25.04 (0.40)
SCPD (two mode)	$\{\widetilde{\mathbf{\Omega}}_{\mathfrak{m},i}\}_{\mathfrak{m}=1}^{2}$	0.12|2.00	0.41|2.00	0.09|2.00	0.40|2.00
	$\{\mathbf{\Omega}^{\star}_{\mathfrak{m},i}\}_{\mathfrak{m}=1}^{2}$	0.05|2.00	0.06|2.00	0.02|2.00	0.14|2.00

to decrease, which is consistent with the theoretical role of $p_{-\mathfrak{m}}$ in amplifying the effective sample size. For example, in Table 1, when $p_1 = p_2 = 60$, the mode-wise standard deviations are essentially zero.

The results also show that SCPD is remarkably robust to initialization. Comparing the rows based on random initialization $\{\widetilde{\boldsymbol{\Omega}}_{\mathfrak{m},i}\}$ with those based on oracle initialization $\{\boldsymbol{\Omega}^{\star}_{\mathfrak{m},i}\}$, the differences are uniformly small across all settings. This observation aligns well with the theory in Section **??**, which only requires the initialization for the remaining mode to lie in a neighborhood of the true parameter and predicts that a single update already yields statistically consistent estimators.

We next consider the joint detection performance. In the synchronous setting **(D1)**, the Hausdorff distance decreases rapidly as the dimensions grow, while the estimated number of change-points remains close to 1, indicating that SCPD correctly identifies the presence of a common change and localizes it precisely. In the asynchronous settings **(D2)** and **(D3)**, the average number of estimated change-points stays at 2.00, and the Hausdorff distance again decreases markedly with dimension. Hence, SCPD is able to distinguish the row-wise and column-wise structural changes instead of collapsing them into a single global event.

By contrast, the vectorized competitors perform substantially worse. Even in the relatively low-dimensional cases of Table 1, hdcd and BA produce much larger Hausdorff errors and severely overestimate the number of change-points. Once the dimensions increase, these methods become numerically unstable or fail entirely. This behavior is expected because vectorization inflates the ambient parameter dimension from mode-specific scales to the full product dimension $p_1 p_2$, while discarding the separable matrix structure. SCPD avoids this curse of dimensionality by estimating the mode-specific precision matrices separately, thereby retaining the structural information and improving both statistical and numerical performance.

Taken together, the three simulation studies reveal two stable trends. First, richer multiway structure leads to more accurate localization, as reflected by the decreasing Hausdorff error with increasing dimensions. Second, even when the sample size is relatively small, SCPD continues to perform well as long as the signal strength is fixed. These observations are fully in line with the theoretical results, which show that the remaining mode contributes an effective sample-size factor and therefore enhances the detectability of structural changes.

6.5 Multiple change point with $\mathfrak{M} = 2$

We further examine the multiple-change-point extension described in Section 5. In all experiments below, we set $n = 100$, $G = 10$, $\nu = 0.4$, and use 100 Monte Carlo replications. Tables 4 and 5 use the same graph structures, dimensions, and values of α as the single-change-point mechanisms **(D1)** and **(D3)**, respectively. In each of these two-mode settings, both the row mode and the column mode contain two true change-points. The synchronous design sets

$$(\tau_1^\star, \tau_2^\star) = \{(30, 70), (30, 70)\},$$

so the row and column changes occur at the same two times and the union of true change-points is $\{30, 70\}$. The asynchronous design sets

$$(\tau_1^\star, \tau_2^\star) = \{(30, 60), (40, 70)\},$$

so the row and column modes each still have two change-points, but they occur at different times and the union of true change-points is $\{30, 40, 60, 70\}$. For these structured graph experiments, the three segments in each mode follow the partial node-permutation design

$$\mathbf{\Omega}^{(A)} \to \mathbf{\Omega}^{(B)} \to \mathbf{\Omega}^{(A)},$$

where $\mathbf{\Omega}^{(B)}$ is obtained by permuting the last α nodes of the baseline graph. Each table entry reports the unnormalized Hausdorff distance between the estimated and true change-point sets, followed by the average number of detected change-points. For the two-mode rows, the estimated set is the union of the row- and column-mode estimates.

Table 6 reports a three-mode tensor experiment. The observations are third-order tensors $\mathbf{X}_i \in \mathbb{R}^{p_1 \times p_2 \times p_3}$ generated from a Kronecker-type tensor normal model by successively multiplying a standard normal tensor along the three modes by the square roots of the mode-specific covariance matrices. The graph types are fixed as hub, band, and Erdos–Renyi for modes 1, 2, and 3, respectively. The tensor experiment considers three single-change-point configurations: the synchronous case has $(\tau_1^\star, \tau_2^\star, \tau_3^\star) = (50, 50, 50)$ and hence true union $\{50\}$; the asynchronous case has $(30, 50, 70)$ and true union $\{30, 50, 70\}$; and the semi-synchronous case has $(40, 40, 70)$ and true union $\{40, 70\}$. The reported SCPD (three mode) rows are formed by taking the union of the three mode-wise estimated change-point sets. For the Erdos–Renyi mode, each regime uses an independently generated random graph with edge probability 0.25 and nonzero symmetric off-diagonal precision entries drawn from $\mathrm{Unif}(0.1, 0.4)$ before enforcing positive definiteness. Because these independently generated random sparse graphs create a stronger and more clearly identifiable signal than the partial node permutations used in **(D1)** and **(D3)**, the tensor ER results in Table 6 are expected to be better than the corresponding structured-graph results in Tables 4 and 5. A compact summary of all multiple-change-point simulation designs is given in Table 7.

Tables 4 and 5 show that the one-pass multiple-change-point procedure performs well for the hub–band mechanisms **(D1)** and **(D3)**. In these two settings, the mode-wise average number of detected change-points is close to the true value 2, and the Hausdorff distances are generally small. For the two-mode summaries, the expected number of distinct change-points is 2 in the synchronous designs and 4 in the asynchronous designs; the reported

Table 4: Multiple-change-point detection performance under data-generating mechanism **(D1)**.

Scenario	Method	Auxiliary	$p_1 = 15(6)$ $p_2 = 15(6)$	$p_1 = 30(8)$ $p_2 = 15(6)$	$p_1 = 30(8)$ $p_2 = 30(8)$
Sync	SCPD (mode-1)	estimated	0.34|2.01	1.30|2.11	0.07|2.00
	SCPD (mode-1)	oracle	0.34|2.01	1.54|2.13	0.03|2.00
	SCPD (mode-2)	estimated	0.25|2.00	0.02|2.00	0.04|2.00
	SCPD (mode-2)	oracle	0.25|2.00	0.03|2.00	0.05|2.00
	SCPD (two mode)	estimated	0.36|2.02	1.18|2.10	0.08|2.01
	SCPD (two mode)	oracle	0.34|2.01	1.20|2.10	0.06|2.00
Async	SCPD (mode-1)	estimated	0.70|2.01	1.22|2.08	0.08|2.00
	SCPD (mode-1)	oracle	0.69|1.99	1.27|2.08	0.08|2.00
	SCPD (mode-2)	estimated	0.23|2.00	0.04|2.00	0.02|2.00
	SCPD (mode-2)	oracle	0.21|2.00	0.02|2.00	0.02|2.00
	SCPD (two mode)	estimated	0.68|4.02	1.12|4.06	0.09|4.00
	SCPD (two mode)	oracle	0.66|4.00	1.15|4.05	0.08|4.00

Table 5: Multiple-change-point detection performance under data-generating mechanism **(D3)**.

Scenario	Method	Auxiliary	$p_1 = 20(8)$ $p_2 = 20(8)$	$p_1 = 40(8)$ $p_2 = 20(8)$	$p_1 = 40(8)$ $p_2 = 40(8)$
Sync	SCPD (mode-1)	estimated	0.10|2.00	2.30|2.11	0.21|2.02
	SCPD (mode-1)	oracle	0.07|2.00	2.42|2.13	0.15|2.01
	SCPD (mode-2)	estimated	0.05|2.00	0.00|2.00	0.05|2.00
	SCPD (mode-2)	oracle	0.05|2.00	0.00|2.00	0.04|2.00
	SCPD (two mode)	estimated	0.14|2.15	1.92|2.57	0.26|2.15
	SCPD (two mode)	oracle	0.11|2.12	2.02|2.54	0.19|2.13
Async	SCPD (mode-1)	estimated	0.11|2.00	1.86|2.11	0.10|2.00
	SCPD (mode-1)	oracle	0.09|2.00	1.90|2.11	0.13|2.00
	SCPD (mode-2)	estimated	0.08|2.00	0.01|2.00	0.03|2.00
	SCPD (mode-2)	oracle	0.10|2.00	0.01|2.00	0.02|2.00
	SCPD (two mode)	estimated	0.19|4.00	1.22|4.11	0.11|4.00
	SCPD (two mode)	oracle	0.19|4.00	1.32|4.11	0.14|4.00

Table 6: Three-mode tensor single-change-point detection performance with an Erdos–Renyi mode.

Scenario	Method	Auxiliary	$p_1 = 20$ $p_2 = 20$ $p_3 = 20$	$p_1 = 40$ $p_2 = 20$ $p_3 = 20$	$p_1 = 40$ $p_2 = 40$ $p_3 = 40$
Sync	SCPD (mode-1)	estimated	0.05|1.00	0.06|1.00	0.06|1.00
	SCPD (mode-1)	oracle	0.04|1.00	0.05|1.00	0.04|1.00
	SCPD (mode-2)	estimated	0.03|1.00	0.02|1.00	0.02|1.00
	SCPD (mode-2)	oracle	0.03|1.00	0.01|1.00	0.02|1.00
	SCPD (mode-3)	estimated	0.04|1.00	0.03|1.00	0.02|1.00
	SCPD (mode-3)	oracle	0.04|1.00	0.02|1.00	0.01|1.00
	SCPD (three mode)	estimated	0.07|1.04	0.07|1.03	0.06|1.03
	SCPD (three mode)	oracle	0.06|1.03	0.05|1.02	0.04|1.02
Async	SCPD (mode-1)	estimated	0.06|1.00	0.06|1.00	0.05|1.00
	SCPD (mode-1)	oracle	0.05|1.00	0.05|1.00	0.05|1.00
	SCPD (mode-2)	estimated	0.04|1.00	0.02|1.00	0.02|1.00
	SCPD (mode-2)	oracle	0.04|1.00	0.02|1.00	0.01|1.00
	SCPD (mode-3)	estimated	0.06|1.00	0.03|1.00	0.02|1.00
	SCPD (mode-3)	oracle	0.05|1.00	0.03|1.00	0.02|1.00
	SCPD (three mode)	estimated	0.10|3.02	0.09|3.01	0.07|3.00
	SCPD (three mode)	oracle	0.09|3.01	0.08|3.01	0.06|3.00
Semi-sync	SCPD (mode-1)	estimated	0.05|1.00	0.05|1.00	0.05|1.00
	SCPD (mode-1)	oracle	0.04|1.00	0.04|1.00	0.04|1.00
	SCPD (mode-2)	estimated	0.04|1.00	0.02|1.00	0.02|1.00
	SCPD (mode-2)	oracle	0.03|1.00	0.02|1.00	0.01|1.00
	SCPD (mode-3)	estimated	0.05|1.00	0.03|1.00	0.02|1.00
	SCPD (mode-3)	oracle	0.05|1.00	0.02|1.00	0.02|1.00
	SCPD (three mode)	estimated	0.09|2.03	0.08|2.02	0.06|2.01
	SCPD (three mode)	oracle	0.08|2.02	0.07|2.01	0.05|2.00

Table 7: Summary of simulation designs for the multiple-change-point experiments.

Reported results	Data type	Graph structure	True change-point setting	Synchrony pattern
Table 4	Two-way matrix	**(D1)**: mode 1 hub, mode 2 band; dimensions and α follow Table 1	Sync: $\mathcal{T}_1^\star = \mathcal{T}_2^\star = \{30, 70\}$; Async: $\mathcal{T}_1^\star = \{30, 60\}$ and $\mathcal{T}_2^\star = \{40, 70\}$	Row and column modes each have two changes; sync means the two modes share both times, while async means the two modes change at different times.
Table 5	Two-way matrix	**(D3)**: mode 1 hub, mode 2 band; dimensions and α follow Table 3	Sync: $\mathcal{T}_1^\star = \mathcal{T}_2^\star = \{30, 70\}$; Async: $\mathcal{T}_1^\star = \{30, 60\}$ and $\mathcal{T}_2^\star = \{40, 70\}$	Same as Table 4; the distinction is the underlying **(D3)** dimension and node-permutation setting.
Table 6	Three-mode tensor	Mode 1 hub, mode 2 band, mode 3 Erdos–Renyi; ER graphs use edge probability 0.25 and independent edge weights from Unif$(0.1, 0.4)$	Sync: $(50, 50, 50)$ with union $\{50\}$; Async: $(30, 50, 70)$ with union $\{30, 50, 70\}$; Semi-sync: $(40, 40, 70)$ with union $\{40, 70\}$	Sync means all modes share one change; async means all modes change at different times; semi-sync means two modes share a change and the remaining mode changes separately.

counts are close to these targets in most settings. The column mode is particularly stable, while the row mode has mild over-detection in the intermediate settings $p_1 = 30, p_2 = 15$ for **(D1)** and $p_1 = 40, p_2 = 20$ for **(D3)**.

Table 6 further demonstrates the tensor extension. The ER component produces a more pronounced graph-level change than the partial node-permutation signals in Tables 4 and 5, so the tensor results are uniformly easier: the mode-wise estimates are tightly localized, and the three-mode summaries recover the correct number of distinct change-points on average across the synchronous, asynchronous, and semi-synchronous scenarios.

7 [Real data applications]

8 Conclusions

This paper developed a framework for change-point detection in Gaussian graphical models with multiway structure. The proposed model allows the precision matrices associated with different modes to change at distinct, coincident, or closely spaced locations. By exploiting the Kronecker structure, SCPD reduces the original joint problem to a collection of mode-specific change-point problems, avoids estimating an unrestricted precision matrix for the vectorized tensor, and permits concurrent updates across modes.

Under the mode-wise AMOC setting, we established theoretical guarantees for change-point localization, precision-matrix estimation, and graph recovery for each mode. We also extended the method to multiple change points through a moving-window procedure followed by segment-wise precision-matrix refitting. Numerical results demonstrate that the proposed procedures provide accurate detection and estimation with favorable computational performance. Future work may develop a full theory for the multiple-change-point extension and accommodate non-Gaussian or temporally dependent tensor observations.

Supplementary material

The supplementary material is provided to support the main article, covering: (i) real-data applications, (ii) additional simulation results and implementation details, (iii) algorithmic pseudocode with explanations, (iv) further theoretical discussions, (v) model assumptions for several examples, and (vi) full technical proofs of all stated results.

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